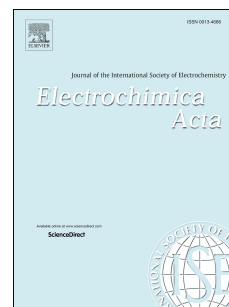


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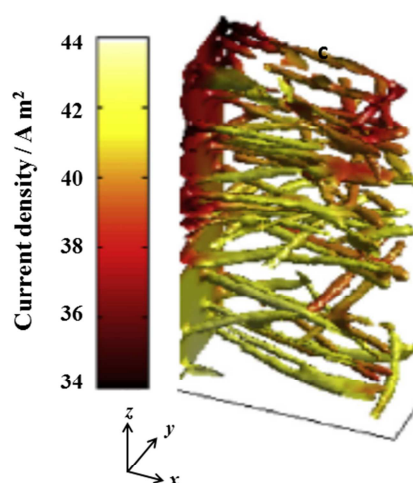
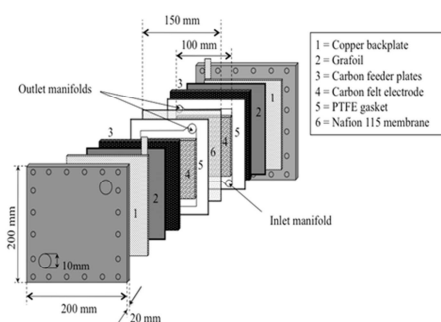
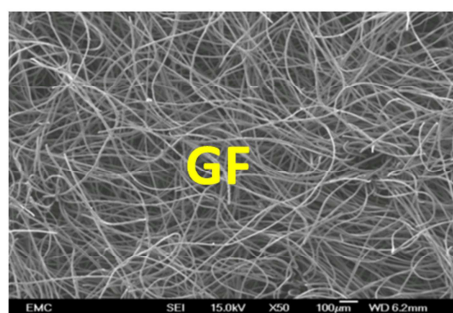
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Graphical abstract

## **Graphite felt as a versatile electrode material: properties, reaction environment, performance and applications**

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**Abstract**

Carbon electrode materials have been extensively studied due to their diverse structure and chemistry which has enabled diverse applications. The thermal synthesis of graphite felt (GF) from polymeric precursors and the physical and electrochemical properties of flexible graphite felts are summarised. In electrochemical flow cells, the three-dimensional structure of GF as an electrode material counteracts the limitations of the low conversion and space–time yield possible with two-dimensional electrodes. The high volumetric area of GF facilitates the treatment of dilute solutions. The physical structure of GF enables it to act as a turbulence promoter, enhancing the hydrodynamics and mass transport rates in electrochemical systems. The electrochemical applications of GF in electroanalysis, electrosynthesis, electrochemical advanced oxidation processes, redox flow batteries, fuel cells, and metal ion removal are illustrated by examples. The characterisation of reaction environment, including non-ideal electrolyte flow, mass transport rates, fluid pressure drop, current and potential distributions of GF in electrochemical reactors are also considered.

**Keywords:** Advanced oxidation of organics; Metal ion removal; Microbial fuel cells; Redox flow batteries; Supercapacitors; Wastewater treatment.

**Nomenclature****Symbols**

|           |                                                     |                             |
|-----------|-----------------------------------------------------|-----------------------------|
| $A$       | Electrode area                                      | $\text{cm}^2$               |
| $A_e$     | Volumetric electrode area                           | $\text{cm}^{-1}$            |
| $c$       | Concentration of electroactive species              | $\text{mol cm}^{-3}$        |
| $c_{(t)}$ | Concentration of electroactive species, at time $t$ | $\text{mol cm}^{-3}$        |
| $c_{(0)}$ | Concentration of electroactive species, initially   | $\text{mol cm}^{-3}$        |
| $d_f$     | Fibre diameter                                      | $\text{cm}$                 |
| $F$       | Faraday constant                                    | $\text{C mol}^{-1}$         |
| $G$       | Electrical conductance                              | $\text{S}$                  |
| $I_L$     | Mass transfer limiting current                      | $\text{A}$                  |
| $j_L$     | Limiting current density                            | $\text{A cm}^{-2}$          |
| $k_m$     | Mass transfer coefficient                           | $\text{cm s}^{-1}$          |
| $L$       | Length of electrode                                 | $\text{cm}$                 |
| $P$       | Fluid pressure                                      | $\text{Pa}$                 |
| $Q$       | Volumetric flow rate of electrolyte                 | $\text{cm}^3 \text{s}^{-1}$ |
| $r_p$     | Pore size                                           | $\text{cm}$                 |
| $T$       | Tortuosity                                          | dimensionless               |
| $v$       | Mean linear flow velocity                           | $\text{cm s}^{-1}$          |
| $V$       | Batch volume of electrolyte                         | $\text{cm}^3$               |
| $V_e$     | Volume of (space occupied by) electrode             | $\text{cm}^3$               |
| $z$       | Electron stoichiometry                              | dimensionless               |

**Greek**

|                         |                                      |                    |
|-------------------------|--------------------------------------|--------------------|
| $\varepsilon$           | Volumetric porosity of the electrode | dimensionless      |
| $\phi$                  | Electrical potential                 | $\text{V}$         |
| $K$                     | Darcy's (liquid) permeability        | $\text{cm}^2$      |
| $K_e$                   | Electrolytic conductivity            | $\text{S cm}^{-1}$ |
| $\rho_e$                | Electrical resistivity of carbon     | $\text{ohm cm}$    |
| $\rho_{e, \text{felt}}$ | Electrical resistivity of felt       | $\text{ohm cm}$    |
| $\rho_g$                | Apparent density of graphite         | $\text{g cm}^{-3}$ |

$\rho_{\text{felt}}$  Apparent density of felt  $\text{g cm}^{-3}$

### Abbreviations

|       |                                                    |
|-------|----------------------------------------------------|
| ADM   | Axial dispersion model                             |
| AO    | Advanced oxidation                                 |
| CF    | Carbon felt                                        |
| CFD   | Computational flow dynamics                        |
| CNF   | Carbon nanofibre                                   |
| EAOP  | Electrochemical advanced oxidation process         |
| EF    | Electro-Fenton                                     |
| EtAQ  | 2-ethyl-9, 10-anthraquinone                        |
| FVM   | Finite volume method                               |
| GF    | Graphite felt                                      |
| LBM   | Lattice Boltzmann method                           |
| MFC   | Microbial fuel cell                                |
| MWCNT | Multi-walled carbon nanotube                       |
| ppi   | Pores per linear inch (as a porosity grade of RVC) |
| PAN   | Polyacrylonitrile                                  |
| PDE   | Plug dispersion exchange                           |
| PEMFC | Proton exchange membrane fuel cell                 |
| PMGF  | Polyaniline-MnO <sub>2</sub> graphite felt         |
| RTD   | Residence time distribution                        |
| RVG   | Reticulated vitreous graphite                      |
| RVC   | Reticulated vitreous carbon                        |
| SSM   | Stainless steel mesh                               |
| TBP   | Tributylphosphate                                  |
| TBAP  | Tetrabutylammonium perchlorate                     |
| VRFB  | Vanadium redox flow battery                        |
| 2-D   | Two-dimensional                                    |
| 3-D   | Three-dimensional                                  |

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## 1. Introduction

Electrochemical technology is playing an increasingly important role in modern society, particularly in the chemical industry and for energy conversion. The key components of all electrochemical cells are the electrodes together with possible separators [1] and turbulence promoters. The electrodes provide the reaction site and in certain cases, such as the processing of dilute solutions, the electrode materials must possess high surface area, suitable porosity, low electronic resistance, and high electrochemical activity for the reactions [2]. Three-dimensional (3-D) electrodes are used to counteract the low space–time yield limitations of electrochemical cells with two-dimensional (2-D) electrodes [3].

Porous 3-D electrodes have been increasingly used for electrochemical processes since the 1970s and a review of such electrodes was published in 1992 by Pletcher *et al.* [4]. Such electrodes can be manufactured from different materials such as metal foams including copper, nickel, zinc, aluminium and stainless steel [5]. Such electrodes can be assembled using expanded metal from stainless steel, titanium, nickel, aluminium and other materials [6, 7]; GF can be used in the uncoated, surface modified or coated condition. Carbon materials that have been employed as 3-D electrodes include reticulated vitreous carbon (RVC), other carbon foams, fibre bundles, woven cloth and graphite felt (GF). Diverse examples of such electrode structures in carbon and metals are provided in Figure 1, advances in 3-D printing providing a versatile additional tool to produce porous electrode structures, e.g. [8].

The advantages and shortcomings of using RVC have been extensively reviewed [9, 10]. GF has a high volumetric surface area,  $A_v$  (22 100 - 22 700 m<sup>2</sup> m<sup>-3</sup>), an electrical



conductivity ( $370.37 \text{ S m}^{-1}$ ) [11] close to that of metals, a high maximum voidage, i.e., volumetric porosity,  $\varepsilon < 0.98$ ), good chemical resistance and stability, favourable liquid permeability,  $K (7.04 \times 10^{-6} \text{ cm}^2)$  after a facile pre-treatment [12], good mechanical integrity, compressibility, ease of manufacture/scale-up, and reasonable costs [13-15]. GF have been employed in applications such as electroanalysis, electrosynthesis, oxidation of organic contaminants, redox flow batteries, fuel cells, microbial cells, metal ion removal and drinking water disinfection, which are illustrated here.

The form of GF needs to be suited to channel flow in electrochemical reactors, commonly of filter press and packed bed assemblies. As indicated in Figure 2, electrodes in rectangular flow channels are typically connected in a flow-by mode, where the electrolyte flow is perpendicular to ionic current lines. In packed beds, the cell is often used in the flow-through mode, where the electrolyte flow is parallel to ionic current flow [5].

Several papers on the experimental characterisation and modelling of reaction environment for different applications on GF fitted in electrochemical reactors have been published in the literature [16]. The hydrodynamics and mass transport characterisation of GF, in addition to the potential and current distributions, are critical, since they play an important role in the selectivity of the desired electrochemical reactions, current efficiency and energy consumption.

A timeline for the development and application of graphite felt electrodes over more than 30 years is shown in the Figure 3. This review considers GF for fundamental research and

pilot-scale industrial applications. The experimental characterisation of reaction environment for various applications of GF, incorporated in flow cells, is discussed. A rigorous analysis of performance data obtained by modelling and simulation recently published in the literature are examined. The physical/chemical properties and manufacture of GF are also reviewed.

The general relationships among structure, properties and applications of GF electrodes are indicated in Figure 4 and specific examples are provided in this review. The manufacture, structure and properties of GF influence its electrochemical characteristics and are outlined in the next section.

## **2. Manufacture, structure and properties**

The diversity of fibre and felt structures of graphite felt and its common polyacrylonitrile, cellulose or Lycra precursors are shown in Figure 5. Lycra was introduced in 1962 by DuPont as a textile combination of polyester and polyurethane polymers. Single PAN (cellulose or lycra) fibres can be graphitised and have found extensive use in environmental detection/monitoring [17] and biochemistry, particularly neurophysiology [18]. Indeed, this technology is well established, such that a number of single barrel replaceable GF and multiple barrel GF dispensers are commercially available. Individual PAN fibres can be bundled or woven to produce an elastomeric textile cloth, rayon being a common brand name. Such textile cloths can be made conductive to enable movement sensing, via dyeing and conductive polymers such as polypyrrole/polythiophenes and/or electroless deposition, as illustrated by Kim *et al.* [19].

## 2.1 Manufacture

GF is manufactured by the carbonisation and graphitisation of the carbon felt, that is a type of carbon fibre, so, in the literature exist different ways to produce the carbon fibres. Figure 5 shows a schematic representation of the main types of precursors to produce GF. The ideal characteristics of precursors required to manufacture carbon fibres are readily converted to carbon fibre with a high carbon yield and via cost-effective processing. Precursors commonly used for carbon fibres are based on textile, cellulosic and pitch materials [20-22].

The main precursor used to produce GF is polyacrylonitrile (PAN), an acrylic fibre [22-24]. GF material from a PAN precursor is normally obtained in six stages: polymerisation, oxidation, carbonisation, graphitisation, surface treatment and washing, then drying sizing and winding, as shown in Figure 5. These steps enable polyacrylonitrile to be converted into a high quality graphite in several hours, depending on the temperature, precursor diameter, and precursor fibre characteristics [21]. During graphitisation, the PAN fibre changes structurally, due to the graphene planes becoming parallel to each other before the graphite structure is finally achieved [22, 25-27].

The cellulosic precursor is normally rayon and the pitch-based precursor is a product of petroleum refining or coal coking. Manufacturing graphite from rayon fibres and pitch based precursor follows a similar procedure to that shown. It is important to note that the physical and chemical properties of the resultant felt depend on the precursor used.

Skyllas-Kazacos *et al.* (1993), demonstrated that the GF from PAN fibre showed a better electrical conductivity and better electrochemical properties than GF from rayon [15], once the GF surface had been activated. In addition, the fibre shape follows the original shape of the precursor used to produce the graphite fibre. There is no evidence of a process where fibres are shaped after carbonisation. These fibres (containing flat ribbons of carbon crystals) are bundled by the manufacturer into thicker fibres then woven into graphite cloth, made into felt, twisted or bundled.

## 2.2 Structure

Many research laboratories use GF as an efficient working electrode (cathode) due to its physical chemistry properties. The GF is normally used in a similar fashion to other porous 3-D electrodes; one of the main advantages of GF stems from its ability to provide a compact reactor for a given duty, together with a high fractional conversion per pass,  $X_A$  [11]. For this reason, it is important to know the intrinsic properties of the porous material such as pore size distribution, volumetric surface area ( $A_e$ ), volumetric porosity ( $\epsilon$ ), and tortuosity ( $T$ ) [11].

The structure of graphite consists of layers of carbon (with covalent and metallic bonding within each layer), the atoms being arranged in a hexagonal pattern within each layer with the layers stacked in an AB sequence (different from the AB sequence in a hexagonal close packed or HCP crystal structure) and are linked by weak Van der Waal's interactions produced by a delocalised  $\pi$ -orbital. This results in a hexagonal unit cell with dimensions  $c = 6.71 \text{ \AA}$  and  $a = 2.46 \text{ \AA}$  [28-30]. The GF shows a filamentary microscopic structure, made

up of fibres of *ca.* 20  $\mu\text{m}$  diameter, exhibiting a high surface area, large void space fraction ( $>90\%$ ), and good electrical conductivity [11].

Figure 6 shows corresponding micrographs of the GF (RVC 4002) taken at various magnifications, where is clearly visible an anisotropic medium constituted by randomly dispersed fibres. The void space between fibres is large (such that a high value of porosity can be expected) and homogeneously distributed, with a geometrical shape quite different from other materials used as 3-D electrodes in Figure 7. The single fibres have a flat, cylinder-like shape, with shallow grooves along the long axis; a more detailed inspection reveals each fibre is formed by a group of thinner ones. A value for the fibre diameter,  $d_f = (19.0 \pm 0.2) \mu\text{m}$  was obtained from mean dimensions in SEM micrographs [11].

Volumetric porosity can be determined using diverse methods including density and modelling of residence time distribution (RTD). Gonzalez-García *et al.* used a density method to find the GF porosity, the pore volume obtained for the range of pore size ( $3.5 < r_p < 15 \text{ nm}$ ) was  $0.7 \text{ cm}^3 \text{ g}^{-1}$  corresponding to a total area of  $176 \text{ m}^2 \text{ g}^{-1}$ . The true density of the felt,  $\rho_{felt}$ , was determined using helium, a monoatomic molecule, small enough to access the lowest size micropores ( $\approx 0.2 \text{ nm}$ ) in the material. The density of  $2.17 \text{ g cm}^{-3}$  is close to that of graphite ( $2.25 \text{ g cm}^{-3}$ ). The value determined for the porosity was of 0.84, calculated from the values of apparent density of the felt,  $\rho_{felt}$  and graphite,  $\rho_g$  [11].

$$\rho_e = \frac{(\rho_g - \rho_{felt})}{\rho_g} \quad (1)$$

In contrast, RTD trials yielded volumetric porosity values higher than 0.97. The concept of average pore diameter for a porous material, made up from a framework of fibres, has no physical meaning but the estimation of this allows establishing the value of void fraction of this material. Moreover, RTD method yields the void fraction of the GF as a function of the thickness ratio, which makes these results very suitable to be used in the design of some aspects of the reactors in electrochemical engineering applications [11].

Carbon felt, paper and cloth electrodes find increasing use in redox flow batteries and fuel cells. Watt-Smith *et al.* have described fractal, gas adsorption, liquid permeametry and electrosorption approaches to porosity and volumetric area determination, quantifying their results with the aid of fractal and other structural models [31, 32].

The value of the specific surface area,  $A_e$  ( $\text{m}^2 \text{m}^{-3}$ ) of a porous material is highly dependent on the method used for its determination. Several methods are reported in the literature, such as: (i) structural methods, such as the filamentary analog procedure, (ii) physical methods:, e.g. physical adsorption of gases such as  $\text{N}_2$ , He and Ar, liquid or gas permeametry, mercury porosimetry and (iii) electrochemical methods. The last techniques include:

a) linear sweep voltammetry with an electrolyte flowing and containing a fixed concentration of an electroactive species,  $c$ , the convective-diffusion limiting current under complete mass transport control, being:

$$I_L = j_L \cdot A = k_m \cdot A_e \cdot V_e \cdot z \cdot F \cdot c \quad (2)$$

$z$  being the stoichiometric electron number and  $F$  the Faraday constant.

b) chronoamperometry, i.e., measurement of limiting current with time with a flowing electrolyte that contains a fixed concentration of electroactive species,  $c$ , the steady state limiting current being measured at a time long enough to achieve steady state but short enough to avoid porosity reduction and surface area changes due to, e.g., metal deposition or carbon degradation. Again, equation (2) can be used to estimate  $A_e$ .

c) controlled potential or constant current concentration decay of an electroactive species contained in a constant volume of electrolyte, under complete mass transport control. The first order decay constant,  $k = k_m A_e V_e$  being related to  $A_e$  by a first order concentration reactant decay,  $c_{(t)}$  from its initial value,  $c_{(0)}$ :

$$c_{(t)} = c_{(0)} \exp (-kt/V) \quad (3)$$

where  $V$  is the constant volume of the electrolyte. These diverse methodologies result in a range of values and the evaluation of the true interfacial area, available for the reaction, is not always well defined. The choice of a suitable method is dependent on the nature of the specific surface area to be quantified, although a comparative study of several techniques might be mandatory. Some authors use the filamentary analog procedure which considers fibres as long, perfect non-porous cylinders. The specific surface area is related to the diameter of the fibre,  $d_f$ , and to the volumetric porosity of the felt,  $\varepsilon$ , by the equation [11]:

$$A_e = \frac{4}{d_f} (1 - \varepsilon) \quad (4)$$

Some values of the specific surface area for GF supplied by different companies are shown in Table 1.

d) cyclic voltammetry to determine monolayer electrosorption of a species, the electrical charge density for adsorption or desorption corresponding to a calibrated amount of species, examples being underpotential deposition/stripping of silver and hydrogen adsorption on platinum coated carbon supports [33].

The tortuosity,  $T$  is a measure of the fluid path length through the porous material related to the actual path length through the carbon felt [34]. The tortuosity values for carbon felt obtained using Ergun method were 4.95, 6.35, 5.07 and 6.04, for different thickness ratios of, 1, 1.125, 1.25, 1.625, respectively and it has been reported that GF shows values typically 5-6 times higher than Ni foam [11].

Compressibility is an important property of the GF. High compressibility makes it possible to add parallel carbon felt sheets to obtain a geometrical volume similar to that offered by a polymer mesh spacer in an electrochemical reactor, without physical damage to the felt. This fact provides several advantages for its application, giving rise to a good electrical connection between the collector and the felt. On the other hand, the permeability depends of the porous media and is defined as the liquid velocity throughout a volume unit per unit of differential pressure. Different permeability values for carbon felt can be found as a function of different average pore diameters, highlighting that the permeability increases with porous media [11] and surface pre-treatments [12].



The structure and properties of the GF have been reported in the literature and the parameters may vary depending on the type of GF chosen, so it is clear the versatility of this electrode in electrochemical applications due to its extensive area, void fraction and permeability.

### 2.3 Chemical and electrical properties

Common carbon electrodes which are chemically inert and have very good electrical conductivity include glassy carbon, GF, and RVC [35]. Graphite felt also provides abundant reaction sites [36].

Graphite is considered as one of the most chemically inert materials, being resistant to most acids, alkalis and corrosive gases. However, impurities are almost always present to some degree and often have an important catalytic effect, which results in an increase in the chemical reactivity. Depending on the degree of porosity, the chemical reactivity of GF can be appreciably affected, since high porosity leads to a large increase in surface area, which increases the overall reactivity. The reactivity increases with temperature. So, chemical reactivity depends on the particular type of graphite material; Table 2 shows a general overview of the chemical resistance of GF [29].

The graphite may be considered as a semi-metal that is an electrical conductor in the basal plane but an insulator normal to the basal plane. The atomic structure is such that the highest-filled valence band overlaps the lowest-empty conduction band by approximately 36 meV. Consequently, graphite is a relatively good conductor of electricity [37]. Some

papers consider GF advantageous as an electrode owing to the high permeability [38, 39].

The electrical resistivity of GF,  $\rho_{e,felt}$  can be calculated as:

$$\rho_{e,felt} = \frac{4}{(1-\varepsilon)} \rho_{e,g} \quad (5)$$

where  $\rho_e$  is the electrical resistivity of the graphite. In the literature, values of electrical resistivity for carbon have a wide range, but all fall between  $4 \times 10^{-5}$  and  $6.5 \times 10^{-5}$  ohm m. The electrical resistivity of GF shows values about 100 times higher than 3-D metals; this is important since ohmic drop within the material cannot be neglected, i.e., from two suppliers, stating values as high as  $7.14 \times 10^{-3}$  and  $2.7 \times 10^{-3}$  ohm m [11]. Gaunand and Coeuret reported values of electrical conductivity of graphite microspheres, used in a fixed bed electrode [40].

As supplied, GF often has poor wettability in aqueous electrolytes and a surface pre-treatment is often necessary to achieve a sufficiently hydrophilic surface. The felts need to be adequately treated to remove air bubbles and achieve a low charge-transfer resistance [41]. Many pre-treatments have been suggested for felt electrodes, including plasma [42, 43], thermal [44-46], electrochemical [47-49] and chemical processes [43, 50-54]. Some authors have activated GF by heating in air and a short thermal treatment can be suited to rapid, mass production. In the case of closely packed fibres and felts, it is often useful to rinse the material in alcohol to help displace entrapped air. Other studies have used boiling in  $1 \text{ mol dm}^{-3}$  NaOH for 1 h to render the fibres hydrophilic, followed by washing with deionized water [55]. There is a lack of clarity concerning changes in surface

functionalisation due to pre-treatment of GF and there are opportunities for further IR and Raman spectroscopy techniques, including *in-situ* studies.

### **3. Electrochemical cell designs for graphite felt electrodes and their characterisation**

#### **3.1 Cell design**

Some authors maintain that GF contributes to the achievement of reduced electrical resistance and increases conversion efficiency. The fibres of the felt are joined together by graphite, increasing electrical conductivity between the fibres and the internal surface area of the felt material, together with the sizes of the interstices. This can achieve unexpected improvements in cell performance. Such properties together with the mechanical flexibility of graphite felt have generated considerable research interest.

Many laboratory-scale undivided cells have been used for experimental studies on porous electrodes [56, 57]. The implementation of 3D electrodes in electrochemical reactors has been widely treated in the past few years [58]. A great diversity of materials has been used as three-dimensional electrodes [59-62] and the scope of their applications is continuously being extended.

One of the main advantages of GF electrodes is that they can provide a more compact reactor for a given duty, together with a high fractional conversion per pass [63]. The properties of graphite felt as an efficient flow-through or flow-by electrode are demonstrated in some applications, i.e., in the removal of traces of mercuric ions from aqueous solutions, where the mass transport and other properties, are much better for GF

than for the other electrode materials, making GF attractive for the use in electrochemical reactors [38].

Electrochemical cells fitted with GF electrodes can involve flow-through and flow-by configurations (Figure 2). The application of flow-through porous electrodes to achieve high conversion in a single pass has been relatively limited. In the flow-through porous electrode configuration the electrolyte flows in a parallel direction to the current. In many cases, a relatively high flow rate through the electrode is used to enhance mass transfer rates, reducing the electrode potential for achieving high conversions. In addition, the effective electrode thickness may be limited by the potential drop in the electrolyte, which can restrict the conversion per pass achieved in practice [64]. This mode of operation has been used in many electrochemical applications [65-70]. Figure 8a) shows an electrochemical cell using a GF electrode in the flow-through configuration, where the GF was supplied from Sigratherm as GFD5; this GF has classically been used to remove silver from spent fixing baths [71]. The scale-up and practical applications of the above arrangement have been limited compared to the 'flow-by' arrangement, where the electrolyte flow is perpendicular to the ionic current lines.

Figure 8b), shows another electrochemical cell (a GBC reactor) in the flow-by configuration. Two RVG 4000 GF electrodes were clamped together with two acrylic frames, separated by a cation exchange membrane (Nafion 115) that divided the cell into two compartments. This flow cell is used to reduce the concentration of hexavalent chromium [72]. This configuration has been popular for some large-scale electrochemical applications, i.e., metal ion removal [73, 74].

The Porocell, shown in Figure 8c) was developed by EA Technology. The cell has a cylindrical casing containing a carbon felt cathode for the electrochemical removal of heavy metals from waste streams. Electrolyte is passed throughout the cell until the electrodeposited metal causes a significant pressure drop, at which time the carbon cartridge is removed and replaced [75]. Coria *et al.* [76], used the FM01-LC reactor, a flow-by cell widely described in literature [16, 77], with a graphite felt electrode as a cathode to study the mass transport during dissolved oxygen reduction to hydrogen peroxide. Pérez *et al.* [78] used an undivided filter-press reactor, with a single electrolyte compartment, using graphite felt in the cell configuration as a cathode, which was attached to a stainless steel plate as current collector, in the study of the oxidation of free cyanide in alkaline media on the 3D Ti-RuO<sub>2</sub> anode. As can be observed, GF can be used in several types of reactor of various dimensions, geometries and configurations.

### 3.2. Characterisation of reaction environment

Since it is well known that the transport variables can affect the electrochemical process the most important tool to analyse the design and performance of electrochemical reactors is the characterisation of the reaction environment. Reasonable mass transport, as well as uniform potential and current distributions, are necessary to enhance the current efficiency. In this context, the mass transport is especially enhanced in systems with GF due to the larger volumetric area, resulting in greater current density values [76]. GF enables a high current efficiency and space-time yield.

Careful consideration of the electrochemical reactor design, containing GF requires rigorous analysis of the reaction environment to achieve a well-engineered flow cell. The variables of fluid flow such as non-ideal flow analysis and pressure drop, mass transport and current distribution must be characterized by CFD simulations and experimental methods. These studies are necessary for the electrochemical process intensification. In order to design an appropriate size of the cell and obtain the optimal operational conditions (such as flow rates, current densities, temperature, initial concentrations) enabling a good conversion with an acceptable current efficiency and low energy consumption during electrolysis.

### 3.2.1 Analysis of non-ideal flow

Depending of the electrode shape and cell geometry, non-ideal electrolyte may appear, affecting pressure drop; on the other hand, GF may also promote electrolyte turbulence within the porous matrix, enhancing mass transport [16].

González-García *et al.* (1998), affirm the interest of the diffusion-convection controlled processes inside GF electrode. These authors studied the behaviour of the electrolyte in a filter-press electrochemical reactor (UA200.08). They performed experiments of RTD for different felt thickness, obtaining a similar curve behaviour independently of the electrolyte flow rate [79]. Two peaks were visible in the RTD, the former being attributed to the formation of two electrolyte flow routes and the second one associated to a recirculation of the electrolyte inside the reactor, caused by a reduction of the transversal area at the exit from the cell.

Cao *et al.* (2011) show RTD experiments that were carried out to study the characteristics of the flow in the composite packing with different loadings of carbon nanofibres (CNFs) [80]. These experiments used cyclohexane as the electrolyte and o-nitrophenol as a tracer. The results of the RTD shows remarkably loading changes attributed to different porosity volumes of the electrode. Moreover, at a higher CNF loading, more electrolyte was trapped in the CNF layer, which provoked a tailing effect in the RTD curve. The tailing is associated to low velocity zones being more prominent at low flows. These authors used a plug dispersion exchange (PDE) model, obtaining good agreement between experimental data and simulations [80]. These authors also studied the axial dispersion in carbon-nanofibre/graphite felt composite, synthesized by growing of the carbon nanofibres (CNFs) on the surface of GF [81]. The results obtained of RTD experiments in Figure 9a) were fitted using the axial dispersion (ADM) and PDE models. The RTD curves shows tailing, associated to low velocity zone, similar to that obtained in [80], which decreased with electrolyte flow rate. Likewise, the Figure 9b) shows the axial dispersion coefficient ( $D_{ax}$ ) values, evaluated from the Figure 9a), which varied linearly with electrolyte flow rate, indicating that this 3D electrode generates turbulence [81]. The values of  $D_{ax}$  were of the order of magnitude of  $10^{-5}$ , which indicates that the flow tend to piston flow.

A knowledge of the flow pattern developed inside GF electrodes is important, since it allows optimal electrolyte flow rates, avoiding stagnant and low velocity zones, which typically result in parasitic reactions, i.e. the consumption of background electrolyte provokes dramatic interfacial pH changes that diminish the current efficiency. Despite the widespread use of computational fluid dynamics (CFD) to simulate flow in filter-press type reactors with parallel plates electrodes [16], the modelling of GF is still rather limited. The

CFD approach provides an opportunity to clarify the local fluid flow behaviour within porous electrodes and selected examples of CFD simulations are presented below.

### 3.2.2 Pressure drop

It is well known that sintered metal, metal foam and GF as electrodes may be suitable for achieving high conversion in a single pass; however, in the electrochemical flow cells a high pressure drop can provoke kinetic energy loss and high pump requirements [65, 68].

Bae *et al.* evaluated GF electrodes of relatively high porosity (95%), finding that a high surface area resulted in a significantly higher pressure drop (estimated from the Ergun equation). The high porosity of the GF materials lead to achieve high reactant conversion [82]. On the other hand, Cao *et al.* (2010) studied carbon nanofibres (CNFs) grown on graphite fibre felt resulting in a composite, which combined the structural properties of graphite fibre, felt and CNFs [83]. These authors measured different pressure drops of the composite and reported that the CNF loading on the composite had a strong influence on the pressure drop, because the CNFs had expanded the diameter of the fibres and increased the roughness and porosity of the composite. These authors also determined the pressure drop over the CNF electrode using the Ergun equation, which allowed an effective prediction from the structural parameters related to the CNF loading, such as macropore porosity and expanded diameter of the graphite fibres [83]. An extended Ergun equation for the composite can be written as:

$$\frac{\Delta P}{L} = 204.8 \frac{(1-\varepsilon)^2}{\varepsilon^3 d_f^2} \mu v + 4b \frac{(1-\varepsilon)}{\varepsilon^3 d_f} \rho v^2 \quad (6)$$



where,  $\Delta P/L$  is the pressure drop per unit length,  $\mu$  is the dynamic viscosity and  $\rho$  is the fluid density of the electrolyte;  $\varepsilon$  is the macroporosity and  $d_f$  the expanded fibre diameter of the composite.

Arenas *et al.* [84] used the Ergun equation to describe the pressure drop over a porous electrode as a function of mean linear flow rate and affirm that the performance of the electrodes can be associated with hydraulic loss through them. The pressure drop through electrode materials such as felts can be reduced by shortening the effective flow path length, using flow-across cell designs.

In 2011, Cao *et al.* carried out pressure drop experiments with ethanol solution to study the influence of the liquid surface tension on the permeability of the carbon-nanofibre/graphite felt composite [80]. The pressure drop of the CNF composite for single-phase flow (gas or wettability liquid) was correlated to the structural properties (i.e., the volume fraction of the large pores and the diameter of the expanded fibres) by means of the extended Ergun equation [80]. Owing to the shrinkage of the CNF layer induced by capillary effects, the pressure drop of the composite can significantly decrease if it undergoes wetting by cyclohexane followed by drying in air. The results showed that the pressure drop over the composite was caused only by the viscous resistance of the liquid, as shown in Figure 10a), which shows the linear dependence of the pressure drop of the CNF composite on the superficial velocity of the ethanol solution. On the other hand, Figure 10b) shows changes in pressure drop and permeability coefficient for ethanol flow and the drying effect on the

CNF composite. The pressure drop decreased while the permeability increased remarkably because of the shrinkage of the CNF layer [80].

### 3.2.3 Mass transport

Mass transport studies have been applied to carbon and graphite felts [38, 61, 85-87]. These felts have been characterized by their smallest dimension, with a fibre diameter of 9 or 10  $\mu\text{m}$ , which results in a very high electrode area per unit volume. The high volumetric area facilitates a good space time yield in a single pass, even with very thin flow-through electrodes [88].

The concentration profile of the electroactive species along a GF electrode, under mass transport control, in a continuous single pass reactor, in the absence of phase changes and dispersion effects, can be described by the ideal PFR equation [61, 89, 90]:

$$c(z) = c(z=0) \exp \left[ -\frac{k_m A_e (1-\varepsilon)}{uv} z \right] \quad (7)$$

Where  $c(z=0)$  and  $c(z)$  are the concentrations of the electroactive species at the inlet of the packed electrode at  $z = 0$  and at a  $z$  point along the electrode length, respectively.  $k_m$  is the average mass transport coefficient, which is assumed to be independent of axial position,  $A_e$  is the volumetric surface electrode area, i.e. electrode area per unit volume occupied by the electrode),  $\varepsilon$  is the electrode porosity and  $v$  is the mean linear electrolyte velocity in an empty channel of the same dimensions. The concentration profile demands the mass

transport coefficient,  $k_m$  which is correlated with mean linear electrolyte velocity,  $v$  via an empirical power law:

$$k_m A_e = b v^c \quad (8)$$

The coefficient  $b$  from equation (8) is associated with the shape and cell dimensions, while the constant,  $c$ , is associated with the hydrodynamics. Nava *et al.* (2009) performed a comparative mass transport study of GF and reticulated vitreous carbon (RVC) as cathodes in a flow-through reactor operating in a single pass mode of operation, during the cathodic reduction of cupric ions [5]. These authors reported that the magnitude of  $b$  for the GF was 3.8 times higher than the obtained for RVC, which is associated to the electrode area (Table 3). The coefficient  $b$ , for the GF was about 10 times higher than that obtained for RVC (Table 3), indicating that within GF there exists more turbulence as expected. A comparison of mass transport coefficient between GF and RVC is shown in Figure 10c).

The mass transport controlled reduction of ferricyanide ions at siliconised GF (99.9%wt purity, 1 cm thick, Morgan Carbon, Luxemburg) in an aqueous solution of potassium ferricyanide has been studied [91], using a sol-gel technology to modify the felt. The gel was formed within internal pores and consisted of nanoparticles with an average diameter of 12 nm. The siliconisation of graphite felt did not affect mass transport of the electroactive anion but significantly slows the mass transport rate of cations. Several reports have considered mass transport in GF electrodes [87, 92-95].

Mass transport rates within GF electrodes are higher than in other planar or three-dimensional electrode types. Not only do the carbon fibres that constitute the felt induce turbulence in the electrolyte, enhancing mass transport but the extensive volumetric surface electrode area of the GF also increases the volumetric mass transport coefficient, i.e., the product of mass transfer coefficient and the area of electrode material per unit volume ( $k_m A_e$ ). This improves the conversion of reactant during bulk electrolysis and the space-time yield. Several experimental studies mass transport have been reported but CFD simulations of mass transport are scarce.

### 3.2.4 Potential and current distributions

The potential drop within GF, in a highly conducting flow, depends on the active thickness of the electrode, the kinetic control of the electrochemical reaction (i.e. charge transfer or mass transport control), concentration distribution and hydrodynamic effects. For an anodic process operating under mass transport limited conditions, the electrode potential can be described by [69]:

$$\Delta\phi = \frac{zFvc_0}{\alpha\kappa} [1 - (1 + \alpha L)\exp(-\alpha L)] \quad (9)$$

$L$  is the thickness of the porous electrode,  $v$ , is the linear electrolyte velocity past the electrode surface,  $c_0$  is the inlet concentration,  $\kappa$  is the electrolytic conductivity and  $\alpha = k_m A_e / v$ , where  $A_e$ , is the volumetric electrode area. Under conditions of high conversion ( $\alpha \rightarrow \infty$ ) equation (9) can be simplified to:

$$\Delta\phi_{mx} = \frac{zFvc_0}{\alpha\kappa} \quad (10)$$

Some authors consider that the conversion of the electroactive species can be increased up to a limit of around 95%, prior to oxidation of the background electrolyte [82]. High conversions and an increasing  $c_0$  result in higher potential drops and current densities.

Nava *et al.*, in (2009) quantified the experimental potential distribution in a flow through reactor using GF and 100 ppi RVC electrodes (1.2 cm thick), during the deposition of copper in the absence of hydrogen evolution in Figure 10d) [5]. Figure 10d) shows that the potential drop remains constant over RVC, meanwhile with GF the absolute potential drop increases with flow rate. The authors tried to fit the results of GF and RVC using a model reported in the literature [90, 96] but the experimental results were an order of magnitude higher than values predicted by simulations. These differences were attributed to a nonuniform potential distribution within the graphite matrix.

The current distribution in a positive current collector of Na/Se(IV) using GF in a radial shape has been measured [97]; the radial potential distribution depended on the arrangement of the positive current collector. Other papers that show experimental results of potential distribution in GF can be consulted elsewhere [35, 98-100].

### 3.2.5 Modelling and simulation

In addition to experimental investigations, the modelling and simulation of hydrodynamics, mass transport and current distribution can also play an important role in the design and

optimisation of electrochemical reactors equipped with GF electrodes [11, 69, 98, 101-107].

Doherty *et al.* performed a mathematical model for an electrochemical reactor equipped with GF in the flow-through configuration [108]. This model presents simulations of the concentration, and potential distributions, highlighting that authors solved the conservation equations, in one dimension and in the steady state, by an analytical method and by finite differences (i.e. by a numerical method) [108] Figure 11 shows a normalised concentration profile along the porous electrode, which compared the plot obtained from the numerical simulation and the exponential profile, described by:

$$c = c_{out} \exp\left(\frac{A_e k_m}{v} x\right) \quad (11)$$

Good agreement between numerical simulations and analytical approach is seen in Figure 11). This is particularly true for GF electrodes with an electrode thickness of 20 mm. The performance of GF electrode in flow-through mode depends on variables such as the electrolyte concentration, electrode material properties (porosity, specific surface area and conductivity), electrolyte conductivity and liquid flow rate. Other researchers have used parametric models to investigate the behaviour of the GF as electrode material [95, 99].

Qiu and colleagues [109] modelled the electrolyte flow rate, concentration of redox species and the charge and discharge in a vanadium redox flow batteries (VRFB) equipped with GF as electrodes. The electrode geometry was obtained using X-ray computed tomography (XCT) analysis and the obtained data were used as geometry input for modelling of the

VRFB. The following Navier-Stokes, convection-diffusion, and Ohm's Law equations were used for modelling the VRFB.

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{v} = -\frac{1}{\rho} \nabla p + \nu \nabla^2 \mathbf{v} \quad (12)$$

$$\nabla \cdot \mathbf{v} = 0 \quad (13)$$

$$\frac{\partial C_j}{\partial t} + \mathbf{v} \cdot \nabla C_j = D_j \nabla^2 C_j + \nabla \cdot \left[ \frac{z_j C_j D_j}{RT} \nabla \phi \right] \quad (14)$$

$$\mathbf{J} = -k_s \nabla \phi \quad (15)$$

where  $\mathbf{v}$  is the electrolyte velocity,  $\rho$  is the electrolyte density,  $p$  is the pressure,  $\nu$  is the kinematic viscosity of the electrolyte,  $D_j$  is the diffusion coefficient of species  $j$ ,  $z_j$  is the charge of species  $j$ ,  $R$  the universal gas constant,  $T$  the temperature,  $\phi$  the electrical potential and  $k_s$  the electrical conductivity of GF. In this study, the authors employed the lattice Boltzmann method (LBM) for the solution of the hydrodynamic problem and the finite volume method (FVM) for the mass and charge transport. The cell performance was investigated as a function of the electrolyte flow rate and external drawing current provided the cell potential distribution, local concentration, and overpotential and current density profiles under galvanostatic discharge conditions [109]. For example, Figure 12a), b) and c) show the local concentration distribution, overpotential and current densities in the negative half-cell. In Figure 12a) a significant depletion of concentration is seen in the  $z$ -coordinate, which is characteristic of porous electrodes. In Figures 12b) and c), the overpotential and

current density distribution is significant in the  $x$ ,  $y$ , and  $z$ -coordinates. The model developed by Qiu *et al.* showed the use of CFD simulations in optimising the performance of the VRFB. Wu *et al.* designed a flow frame with a multi-distribution channel, simulating the electrolyte flow to be distributed inside the GF electrode of a VRFB using CFD techniques [110].

Watt-Smith *et al.* [111] used a mathematical model to describe the charge and discharge performance of a VRFB, obtaining a robust model for the performance characteristics over multiple cycles of charge and discharge, the authors demonstrated via cyclic voltammetry that the GF (GFA type materials provided Sigratherm<sup>®</sup>) exhibited the best performance of a wide variety of carbon felts considered.

Kear *et al.* [112] reviewed technological, financial and policy aspects about the development of the all-vanadium redox flow battery for energy storage, where the authors refer to modelling studies of redox flow batteries. The issues that have been addressed using modeling together with the current and future requirements of modelling are outlined. Different works report the VRFB modelling using carbon electrodes [113-116].

Numerical simulations can be used as a first approach to predict the cell performance, enabling rapid testing of valuable information inside the GF. The accuracy of the simulation depends not only on the rigor and robustness of the model formulation, but also on the accuracy of transport properties and parameters of the electrolyte and electrode matrix employed as inputs to the model. It is worth mentioning that in spite of the advantages of modelling, there are only few CFD simulations in GF that have not been



focus on RFB; these numerical approaches can address other electrochemical systems and applications.

#### **4. Applications**

The increasing use of electrochemical technologies in a variety of applications such as synthesis, energy storage and environmental treatment, is due, among other reasons, to the utilisation of porous materials as three-dimensional electrodes in the design of electrochemical reactors. In this context, the GF has been used in several electrochemical applications, such as electroanalysis, electrosynthesis, environmental treatment and, energy conversion and storage.

##### **4.1 Electroanalysis and electrode kinetics**

Electrochemical reactions on carbon-based electrodes are normally slower than at noble metal electrodes, in this way, the electron transfer kinetics depends on surface preparation and structure, and some background current, making a comparison with metallic electrodes, the carbon-based electrodes have a wider potential range for the electrolysis of water. These carbonaceous materials are generally low cost and are chemically inert [117].

Continuous flow analytical systems are advantageous because are easily automatable and lead to obtain real-time analysis. Electrochemical flow detection has been the subject of investigations in many areas. In electrochemical sensors, where key is the limit of detection, it is necessary to maximize the current intensity to increase the limit of detection, this later is achieved with the extensive area of GF [118].

Many GF sensors have used flow cells in which controlled hydrodynamics increase the mass transport [119-121]. The GF electrodes are “mega-macro” hybrids that have been used for electroanalysis at very low levels ( $<1 \mu\text{g dm}^{-3}$ ) of concentration [122, 123].

Several electroanalytical techniques are used to detect a large range of metallic ions and important organic/organometallic molecules at low concentration, Stripping voltammetry for example, used by some authors, employs a GF electrode [118, 122]. Geneste *et al.* used GF flow cells to detect a number of metal ions, including lead copper and zinc, with impressive results, suggesting the “mega-macro” electrode concept. Several researchers have used hybrid mega-macro carbon electrodes for anodic stripping voltammetry (ASV) technique in quiescent solutions for the determinations of heavy metals [124, 125]. Table 4 provides some examples of recent sensor studies using GF electrodes; the determined species include silver [117, 126], zinc [127], lead [128, 129], hydrazine [130], nitrite [131] and hydrogen peroxide [132].

In some studies the optimisation of the detection limit is improved by means of modification of GF [129], i.e. modified electrodes have shown the ability to detect  $\text{Pb}^{2+}$  ions with no interference from  $\text{Cu(II)}$ . GF modification with the covalent immobilisation of cyclam shows good differences compared with an unmodified GF electrode. GF has a typical surface area 800 times greater than pyrolytic carbon (*ca.*  $55 \text{ cm}^2$  compared to  $0.07 \text{ cm}^2$ ), although the ratio of the peak currents between GF and pyrolytic carbon is *ca.* 30 [122].

The use of the graphite felt in electroanalysis is common, allowing detection of ions with and without modification of the GF under static and flowing electrolyte conditions [128, 129].

## 4.2 Electrosynthesis

Electrochemical synthesis covers the production of inorganic and organic chemicals, many electrochemical syntheses involve a change of phase during the formation of the product. In addition, the electrosynthesis simplifies product recovery, separation and it reduces the tendency to generate waste streams [133]. In 1990, Couper *et al.* reviewed electrode materials for small-scale electrosynthesis processes; the structure of carbon electrodes allow the intercalation or impregnation by inorganic and organic materials [134], i.e., the intercalation of anions [135] and cations [136] was reported.

### 4.2.1 Inorganics

The GF is used to the electrochemical synthesis of Cr(III) [137]. In a potential range of -0.8 to 0.8 V vs. SCE, the electrochemical reaction at carbon electrodes is essentially a surface process of proton adsorption and desorption, without Cr(II) formation and significant hydrogen evolution. But at more negative electrode potentials than -0.8 V vs. SCE, both hydrogen evolution and chromium (II) formation occurred simultaneously. The electrochemical reduction of Cr(III) on GF electrodes is controlled mainly by charge transfer rather than mass transport, at electrode potentials of -0.8 V to -1.2 V vs. SCE.

The electrosynthesis of hydrogen peroxide in acidic solutions (pH 3) with dissolved oxygen takes place in GF and RVC electrodes, fitted into a filter-press electrolyzers, at superficial

current densities up to  $3000 \text{ A m}^{-2}$  [138]. Figure 13 shows a comparison between GF and RVC cathodes in a flow-by cell for electrosynthesis of  $\text{H}_2\text{O}_2$ , where the  $\text{H}_2\text{O}_2$  concentration per pass and current efficiency were generally higher for graphite felt than for the RVC electrodes owing to their higher area.

The GF (GF-S4) is used as promising anodic material to the transformation of sulfur dioxide into either sulfuric acid or colloidal sulphur, on the anode or cathode, respectively [139], using lower values of anode potential than 1.5 V vs. SCE, where the main anodic reaction in a GF electrode is the formation of sulfuric acid. In this case the electrochemical reactor with a rotating cylinder electrode allowed to obtain an appropriate performance for the conversion of sulfur dioxide into useful chemicals. The anodic oxidation of Mn(II) to  $\text{MnO}_2$  at graphite electrodes, is an effective application of GF [140], and comparing the GF against metal electrodes, on this application, the carbon material being superior to the others.

Many papers have reported the use of GF for inorganic electrosynthesis. The filter-press type reactor configuration equipped with GF electrodes allow facile scale up, making such processes attractive for industrial applications.

#### 4.2.2 Organics

GF was used to compare the efficiency of electrolysis cells of different sizes [68], in flow-through mode for the electrosynthesis of 3-nitrosophthalic acid from 3-nitrophthalic in aqueous acetic buffer. The GF have good chemical and electrical yields, and in the

preparation of nitrosobenzenes the GF has been shown to be an efficient electrode, because this type of compounds are difficult to obtain using classical chemicals.

Another organic applications are the electrochemical synthesis of heterocyclic compounds through the anodic oxidation of 2'-hidroxichalcones, using graphite felt electrodes as anode with controlled potentials in a divided cell [141]. The electrochemical conversion of  $\alpha$ -nitrobenzenoid compounds into the corresponding oximes by means of an cathodic reduction at graphite felt was performed in aqueous acetic buffer (65%) and ethanol (35%) [142]. The simultaneous electrosynthesis of furoic acid and furfuryl alcohol from furfural, using a GF electrode modified in a separated flow cell is another example [65]. In these applications, GF allow conversion yields up to 90%, with high selectivity and chemical yields, making GF a good option for organic electrosynthesis.

Several papers have reported the use of GF and modified GF for organic electrosynthesis [143-150]. These electrodes enabled diverse organic compounds to be produced with higher selectivity, low cost of electrolytic energy consumption, and the most important, electrosynthesis avoids the use of separation process of difficult mixtures minimizing the discharge of a toxic by-products.

#### 4.3 Electrodeposition

Figure 14 shows tin-copper alloy deposits on reticulated vitreous carbon, RVC and on a porous carbon felt [151]. Figure 14a) shows the surface microstructure and Figure 14b) the surface alloy content, corresponding to a deposit of 4Sn46Cu50Bi. Figure 14b) shows a cross-sectional image of the tin-copper alloy

on the carbon felt substrate shown in Figure 14c). Figure 14d) shows the uniformly deposited alloy on the surface of a single string of the carbon felt. Investigations can be carried out to understand the growth of deposits (layer, columnar) on the substrate. Further studies could be carried out on deposition of composite and multilayers coatings on both the reticulated vitreous carbon (RVC) and carbon felt surface. The former support material has the benefits of flexibility and high surface area porosity but is limited by microporosity which changes with compression as well as surface chemistry (which is easily altered by adsorption). The RVC structure presents a rigid, open macropore network which provides a low pressure drop for electrolyte flow but suffers from being brittle in applications such as batteries and fuel cells.

#### **4.4 Environmental treatment**

GF electrodes have been widely used for the prevention and remediation of environmental pollution, especially focusing on water streams, where new advanced oxidation processes based principally on Fenton like reactions have attracted the attention of environmental electrochemists [152, 153]. In this context, the so-called advanced oxidation processes (AOPs) including heterogeneous processes like anodic oxidation and photoelectrocatalysis are effective to destroy persistent organic pollutants yielding, in certain cases, complete mineralisation of organic pollutants [154-158]. Moreover, metal ion removal from waste water is an established approach to clean polluted water for reuse, while allowing the recovery of the metals in their zero oxidation state making this a sustainable process [159, 160].

#### 4.4.1 Advanced oxidation processes

The electrochemical advanced oxidations processes (EAOPs) such as anodic oxidation (AO), electro-Fenton (EF), photo electro-Fenton (PEF) and solar-photo-Fenton reactions typically deploy a GF anode to induce direct oxidation of organics, or as cathode to produce  $\text{H}_2\text{O}_2$  from acidic solutions in the presence of oxygen [161]. The  $\text{H}_2\text{O}_2$  is employed for Fenton-like reactions, small amounts of  $\text{Fe}^{2+}$  being added as catalyst [152]. Fenton oxidations are promising for the incineration of organic pollutants, based on the well-known hydroxyl radical mechanism [76, 162-164].

In this context, GF has been used for the degradation of organic pollutants in a rotating disk reactor [165], using the methyl orange dye as a model pollutant and dual rotating GF disks to supply oxygen instead of the traditional aeration method, reaching 58.7% of total organic removal. Another example is degradation of sulfamethazine (SMT) by EF process, testing platinum (Pt), boron doped diamond (BDD), dimensionally stable anode (DSA<sup>®</sup>, Ti/RuO<sub>2</sub>-IrO<sub>2</sub>) and graphite felt (GF) as anodes, and GF as cathode. Figure 15a) and b), show the effect of current density on the kinetics of degradation of SMT using the GF/GF electrode couple for EF and anodic oxidation, respectively, where it can be seen that the antibiotic concentration decay is very fast; at 15 minutes the pollutant is almost degraded, which was 3-4 times shorter than the other electrodes tested [166]. The advantage of GF over other electrode materials is the volumetric area that allows a high performance for the removal of the pollutants, besides being a lower cost material than Pt or DSA<sup>®</sup> coated electrodes.

In another study, the degradation of *p*-nitrophenol was carried out using a GF chemically modified as a cathode in an EF process. The modification of GF was performed with

ethanol and hydrazine hydrate to increase their electrocatalytic performance for the  $O_2$  reduction to yield  $H_2O_2$  (giving accumulations from  $67.5 \text{ mg dm}^{-3}$  with an unmodified GF to  $175.8 \text{ mg dm}^{-3}$  using a modified GF); so, a more efficient degradation of the *p*-nitrophenol was achieved with modified GF [167]. The use of GF as cathode in the EF process are very used and reported in various studies, where this porous material is a good option for the *in-situ* production of  $H_2O_2$ . Surface modification of the GF enhances performance, shortens the process time and enables a low electrolytic energy consumption.

#### 4.4.2 Metal ion removal from wastewater

Industrial wastewater streams containing heavy metals represents a serious pollution problem for conventional wastewater plants and to the natural environment, because these metals are toxic for microorganisms founding in sewage plants and for the aquatic life in surface water bodies. In this context, the metallic finishing industry and abandoned mines are sources of such polluted water [72, 168-173]. Different metal ions can be removed for different solutions using porous materials like GF electrodes [75, 174] among the metal ions are found Pb(II) [103], Cr(VI) [72, 175, 176], Cu(II) [100, 173, 177], Zn(II) [173], Hg(I) [38], Ag(I) [71, 178], Au(I) [179, 180].

Several heavy metals are susceptible to be deposited, in the metallic form, on the cathode surface. The zero oxidation state is the most valuable form for several metals. Some studies mentioned the convenience of using 3D carbonaceous cathodes [181] such as reticulated vitreous carbon [63] and GF [182]. Such 3D electrodes have a high volumetric area which is desirable for removal of heavy metals from dilute solutions (having electroactive species concentrations below  $50 \text{ mg dm}^{-3}$ ). For example, GF electrode has



been used in deoxygenated solution to remove copper and zinc ions from diluted solutions [173]. Figure 16 shows the removal of cupric ions from synthetic drainage water and it is seen that 99.9% of the initial concentration ( $150 \text{ mg dm}^{-3}$ ) is removed with a current efficiency of 80-90% at a current density of  $5 \text{ mA cm}^{-2}$ , at higher current densities ( $>5 \text{ mA cm}^{-2}$ ) the current efficiency decreases because of the apparition of the parasitic hydrogen evolution reaction. The removal of zinc ions achieved conversions of 99.5% (from initially concentrations of  $200 \text{ mg dm}^{-3}$ ) with current efficiencies of 65%. It is worth mentioning that during zinc deposition, the simultaneous hydrogen evolution reaction take place, which impact on the current efficiency and electrolytic energy consumptions.

Another example is the removal of gold, with conversions higher than 90% and current efficiencies of 6% to 12%, from solutions containing  $30 \text{ mg dm}^{-3} \text{ Au(I)}$  [179]. Oren and Soffer demonstrated the good properties of graphite felt as an efficient flow-through electrode for removal of mercuric ions, obtaining removal above 99% [38]. GF electrodes have also been used for the removal of  $\text{Cr(VI)}$  ions from wastewater and synthetic electrolytes [176], as well as the recovery of silver from a spent photographic fixing solution in a flow-through cell, obtaining a ca. 90.7% conversion [71].

A pre-pilot flow-through reactor using several GF cathodes, in a cascade arrangement, was used to remove copper from spent baths from the electroplating industry [5]. This well-engineered cell allowed the efficient removal of copper avoiding hydrogen evolution. Another example is the Porocell (flow-across reactor) [75], which has been used to remove copper from wastewater from the whisky distilling industry.

GF is an attractive electrode for industrial applications for the recovery of metal ions from diluted solutions since it possesses a large volumetric surface area and is often available in large rolls. Moreover, the liquid flow inside the porous matrix can become turbulent, providing high mass transport rates towards the electrode surface. The flow-through and flow-across configurations permit acceptable current efficiencies, avoiding hydrogen evolution, the small thickness of GF ( $< 2.4$  cm) allowing a uniform current distribution.

#### **4.5 Energy conversion and storage**

Nowadays, the electrochemical energy conversion and portable storage technologies for long time with high energy density have become a reality for their application in electronic devices [183, 184]. Here, carbon materials have definitively been playing a significant role in the development of alternative clean and sustainable energy technologies [185]. Graphite felt is a typical electrocatalyst support in various electrochemical energy conversion and storage devices [186], such as batteries, redox flow batteries, fuel cells, microbial fuel cells and supercapacitors. Cai *et al.* [187] reviewed the application of carbon fibres in energy conversion and storage devices, where a variety of carbon fibre (CF) materials, such as chopped CFs, carbon cloths/felts, CF textiles and carbon nanofibres, have attracted considerable attention.

##### **4.5.1 Batteries**

Carbonaceous materials plays a particularly important role in the development of Li-ion batteries, due to the reversibly reaction with lithium [185, 186, 188]. To achieve the energy

density of these batteries, carbon materials like graphite felts have been tested for this application [188, 189].

The use of graphite in rechargeable batteries has been proposed a long time ago [190], the first example of lithium storage in a carbon-based material considered was the lithium-intercalated graphite [191]. In this context, lithium intercalated graphite has been proposed as cathode by Armand and Touzain, this cathode material must be rechargeable and exhibit structural integrity during both the charging and discharging cycles, besides to maximize the voltage values [192, 193]. remains the main anode support material for commercial Li-ion batteries and the topic has been extensively reviewed [191].

The intercalated compounds are attractive for cathode materials because of their potential for low weight relative to energy density and because the use of an intercalant species for ion exchange addresses the need for a wide range of accessible non-stoichiometries [193].

On the anode-side, graphite is the material of choice for most lithium-ion chemistries. It has a theoretical charge capacity of  $372 \text{ A h kg}^{-1}$  [184, 185, 188, 191, 194, 195] and an observed capacity of  $280\text{-}330 \text{ A h kg}^{-1}$ , depending on the type of graphite. The potential of  $\text{LiC}_6$  against the  $\text{Li/Li}^+$  reference electrode is practically zero; this implies that the open-circuit voltage completely depends on the reduction potential of the coupled positive material. However, this could cause lithium deposition at the negative electrode surface during fast charging (especially at cold temperatures) without proper control [184].

Unfortunately, the intercalation capacity of Li ions in graphite is still limited [185]. On the other hand, soft carbons (heat-treated at 500-1000 °C) give a reversible capacity of nearly 700 mA h g<sup>-1</sup> with a characteristic plateau in discharge/charge curves at *ca.* 1.0 V; hard carbons (*ca.* 1100 °C) give a reversible charge capacity of 600 mA h g<sup>-1</sup>, but with some irreversible capacity loss and polarisation. In the commercial cells, well ordered graphite, such as meso-carbon microbeads heat-treated at 3000 °C, natural graphite and non-graphitizing carbons (hard carbon) have mainly been used [194]. These different anode materials show different outputs with a constant and slightly inclined discharge potential profile in the commercial cell, which is more suitable for cellular phones and EV battery applications, respectively [184]. While there has been an increased interest and attempts to improve stationary storage using graphite, investment in the area is largely confined to the use of graphite felt in Li-ion batteries for portable electronic devices.

There are some alternatives of Li-ion batteries that use carbonaceous materials as electrodes, such as Li-air [196-203] and Li-sulfur batteries [204-211]. In the Li-air battery, the intercalation cathode of the Li-ion cell is replaced by a porous, electronically conducting substrate, usually carbon, flooded with the electrolyte [212]. For example, Xiao et al. [213], used a paper-like functionalized graphene as a Li-air battery electrode obtaining a high capacity of 15000 A h kg<sup>-1</sup>. This finding is attributed to the unique bimodal porous structure of the electrode, which consists of microporous channels facilitating rapid O<sub>2</sub> diffusion, while the highly connected nanoscale pores provide a high density of reactive sites for Li-O<sub>2</sub> reactions [213].

The Li-S batteries have the capability to satisfy the high energy demands of the next generation of batteries. In the Li-S batteries, the  $\text{Li}_2\text{S}$  is used as the cathode active material contained in a porous electrode matrix, usually carbon [212]. Carbon and carbon papers are used as electrical conductors for increasing the cathode conductivity and as a storage container for the active material within its micropores [206-208, 214]. Lin et al. [206] used a graphene-sulfur composite achieving an electrical conductivity of  $1820 \text{ S cm}^{-1}$ , superior to most reported about graphene; in addition, these authors reported a high discharge capacity of  $615 \text{ A h kg}^{-1}$  [206]. Manthiram et al. [207] reviewed the rechargeable lithium-sulfur batteries, emphasizing on the carbonaceous materials (carbon nanofibbers, graphene, carbon nanotubes and carbon composites) as conductivity enhancers in the cathode [207]. Currently, the graphite felt has not been used as a component in the Li- $\text{O}_2$  and Li-S batteries; however, given its characteristics of conductivity, surface area, compressibility and easy handling, this material can be a good candidate for its use in batteries of this type.

#### 4.5.2 Redox flow batteries

The widely material electrodes used in redox flow batteries (RFBs) are the graphitic or vitreous carbon materials [215-219]. PAN-based carbon and GF's are used as electrode in a variety of battery designs including vanadium redox flow batteries (VRFBs).

An ideal electrode material to RFB should provide a high electrical conductivity, good mechanical properties and strong chemical resistance, be reasonably priced, and have a long cycle life in highly oxidizing media. Leung et al. (2012) [220], reviewed RFB applications in energy storage where shows some materials used in RFBs, highlighting that carbon-based electrodes are the more common used than DSA<sup>®</sup> type electrodes, because

the carbon-based do not undergo dissolution or formation of oxide during chemical oxidation [220].

Since the early days of VRFB development [112], GF has been a favoured material. Studies at Southampton have described the performance of a laboratory pilot flow cell using GF electrodes of 100 cm<sup>2</sup> projected area [221]. Having evaluated more than 8 felts, GFA5 was selected for a study of operational parameters including current density, electrolyte flow rate and vanadium concentration, together with electrolyte monitoring of the state of charge (SOC) using visible spectroscopy and the modelling of cell potential difference during charge-discharge cycling.

In a 2013 critical review, Parasuraman *et al.* [222] reviewed materials for vanadium redox flow battery (VRFB) applications, particularly plates, cloths and felts made of graphite, which are used as both positive and negative electrode. The surface of the graphite electrode used as anode present a slow disintegration due to the oxygen evolution during cell overcharge as a result of a slow oxidation. GF exhibits good stability as a cathode [217, 223, 224].

The GF has been used in VRFBs, employing vanadium ions to store energy in both half-cell electrolytes [225]. Table 5 summarises the use of GF in various studies, as anode and cathode, and in diverse RFB systems.

Surface treatments of graphite electrodes are reported to improve the reaction kinetics in RFBs [54, 226-230], among which are chemical etching [231], thermal treatment [232],

chemical doping [233] and addition of metallic catalyst sites to the carbon fibres [234]. Usually GF can be modified by treating it with sulphuric acid followed by heat treatment, increasing the chemical activity [46, 53, 220, 235]. Modified GFs have demonstrated their ability to withstand oxidant environments while maintaining a good electrical conductivity and well electron-transfer surface [230, 236, 237]. In addition to the activation of GFs, it is also important to understand the relevance of the operational variables within the VRFBs, including electrolyte flow, current density, temperature and electrode material, which can improve battery efficiency [111].

A marked trend in the use of GF for VRFB applications is the incorporation of graphene flakes and carbon nanotubes, especially MWCNTs into the electrodes, either as surface impregnations or composites. Recent examples include [238] and [239].

The VRFB incorporating GF electrodes has been improved through the use of advanced electrode materials and electrode design to enhance their performance, i.e., using a GF electrode which had previously received microwave treatment [240]. GF is also used in RFBs using  $\text{Fe}^{\text{II}}/\text{Fe}^{\text{III}}$  and  $\text{V}^{\text{IV}}/\text{V}^{\text{V}}$  couples where the vanadium catholyte is regenerated by oxygen in the presence of nitric acid and the iron anolyte is reduced by oxidation of methane at 120 °C catalysed by Pt [241].

Another type of RFBs that uses GF is the H-VRFB, which shows a good reversibility and density power peaks of 114 mW cm<sup>-2</sup> [242]. A novel GF modified with WO<sub>3</sub>, obtained via a hydrothermal method, facilitated a higher voltage efficiency in the H-VRFB (88.7% vs. 78.9% for untreated graphite at 70 mA cm<sup>-2</sup>) [243].

Studies have reported the use of nanoparticles of copper deposited on the GF in the VRFBs, improving significantly the electrochemical kinetics of the  $V^{3+}/V^{2+}$  redox reaction, obtaining energy efficiency (EE) of 53.1% and 17.8% higher than the battery without the copper as catalyst, as shown in Figure 17, where cycling tests of VRFBs are performed using electrolyte with and without  $0.005 \text{ mol dm}^{-3} \text{ Cu}^{2+}$  ions [244]. The nanoparticles of copper are formed during charging. The use of composites or catalysts on the GF surface enhanced the performance of RFBs that without using the composites or catalysts.

#### 4.5.3 Fuel cells

Graphite is a vital structural component of several types of fuel cell [245]. In the 1960s, the use of graphite for cell construction materials was development, showing sufficiently stability to replace the more expensive gold-plated tantalum cell, which was reviewed by Appleby [246] and Kordesch [247].

In phosphoric acid fuel cells (PAFCs) and proton-exchange membrane fuel cells (PEMFCs), single sheet of graphite, carbon-form graphite and carbon blacks to composite materials have been chosen like components for fuel cells, due to the unique electrical and structural properties that make it an ideal material in the topic [247]. In PEMFCs the machined graphite and graphite/polymer mixtures, has been used for the bipolar flow-field plates and the gas-diffusion layer placed between the membrane-electrode-assembly [248, 249], inclusive graphite can also act as a support for the active metal in the catalyst layer [245].



Kourasi et al. [250] reviewed the incorporation of polyoxometalates (POM), in particular heteropolyacids (HPA) as catalyst, in the membrane of PEMFCs and direct methanol fuel cell (DMFC) in order to improve proton conductivity, reduce methanol crossover and enhance cell performance. PEMFCs and DMFCs cathodes often use carbon material (cloth, foam, paper) as a gas diffusion layer but the use of POM or HPA in a fuel cell utilising GF has not been reported.

The GF has been studied like electrode support in a  $H_2/Br_2$  fuel cell using GF prevenient of PAN precursor, modified by acidic oxidation and thermal oxidation for improving cell performance, enhancing the electrochemical catalytic activity in the GF as shown in Figure 18. Surface modification of the GF allowed enhance the power densities of the  $H_2/Br_2$  fuel cell from  $0.51$  to  $0.69\text{ W cm}^{-2}$  [36]. However, it is well-known that higher power densities in PEM cells can provoke deterioration in the cell performance by the generation of high amounts of oxygen.

In similar studies, the use GF/BP2000 composite electrode allowed an improved maximum power density of  $0.91\text{ W cm}^{-2}$  to  $1.28\text{ W cm}^{-2}$  after heat treatment, where a comparison of the maximum power densities in two similar applications of GF in  $H_2/Br_2$  fuel cell are presented [251]. The difference between these two studies can be attributed to the BP2000 catalyst and Nafion binder used, since the use of composites can improve the performance of the graphite electrodes.

The conventional direct methanol fuel cell (DMFC) can also use GF electrodes. In this case, the GF hi catalysed by electrodeposition of PtRu and PtRuMo, [252]. A comparison

between these two different deposits proved that the PtRuMo catalyst gave a better performance than PtRu.

The type of flow cell configuration to improve the fuel cell performance is of common research, i.e., Bauer et al. used flow-by and flow field configurations, being the flow-by better reaching  $0.2380 \pm 0.1 \text{ W cm}^{-2}$  at a power density of  $0.7 \text{ A cm}^{-2}$  [252]. The authors extended the reaction zone anode for methanol oxidation, to obtain power densities of  $0.22 \text{ W cm}^{-2}$  with GF-PtRuMo, and  $0.19 \text{ W cm}^{-2}$  with GF-PtRu [253].

Several studies using GF and modified GFs electrodes have been proved in different types of fuel cells like PEMFC, DMFC, DBFC, PAFC [36, 245, 248, 251-256]. It is noteworthy that GFs with supported catalysts enhance the performance of the fuel cells to obtain higher power densities. Studies of modified GF using composites, surface treatments, catalysts, and different types of flow configurations continue.

#### 4.5.4 Microbial fuel cells

Several carbonaceous materials including the GF are widely used as electrode effectively suitable in microbial fuel cells (MFC's) [257-259]. In this context, GF is used as a component in both the anode and cathode [260-262]. Govind [258] and Zhou [257] reviewed the use of carbon electrodes in MFC's applications, comparing diverse carbon electrode materials. Osman et al. [263, 264] reviewed recent developments in MFC technology, mentioning developments of MFC's where the GF is used as anode and cathode in the immobilisation of enzymes and bacteria. An earlier review was carried out by Bullen *et al.* [265] which considered literature on enzymatic and microbial biofuel cells

using diverse substrates, including GF, to carry microorganisms, acting as microreactor to provide appropriate conditions enzymatic oxidation of organic matter.

The comparison between the performance of GF rod, felt and foam based on the surface area of the resulting electrode has been studied [266]. The results indicate similar currents and biomasses for the first two electrodes; on the other hand, the graphite foam produced 2.4 times more current density and 2.7 times more current density than the others. These results can be attributed to the superficial area of the electrodes where the microbial consortium can colonize, increasing the power output.

The MFC is also modified with a highly active catalyst, being one of the most popular the Pt [267]. Some researches use GF containing Pt as cathode, enhancing three times the current intensity than that the obtained with GF, reaching a power density of  $0.15 \text{ W m}^{-2}$  [268].

It is well-known that the Pt and black electrodes are better electrode materials than graphite, graphite felt and carbon-cloth electrodes, for both anode and cathode reactions, but their costs are much higher. However, the use GF with supported Pt, enhanced the power of the MFC three times higher than the pure GF, reaching a power density of  $150 \text{ mW m}^{-2}$ . Table 6 shows other examples of the use of GF in MFCs.

Figure 19 shows the performance of power density and cell voltage in MFC for GF, carbon paper (CP) and stainless steel mesh (SSM) used as cathodes [269]. GF is used as biocathode with anaerobic sludge as inoculums to achieve a high maximum current and

power density compared to the other two electrodes; the extensive volumetric area of GF plays an important role in MFC's. It is worth mentioning that the composition of the aqueous medium and the type of microbial colony also influence the performance of the MFC.

#### 4.5.5 Supercapacitors

The attraction of the use of carbon as a supercapacitor electrode material arises from a unique combination of chemical and physical properties, the main characteristics are the high conductivity porous carbon material, high surface area range and flexibility, being the unique three-dimensional mesh-like structure that can provide an easier path for electrolyte-ion penetration, desirable for any type of supercapacitor devices [270, 271].

Modified graphite felts have been widely used as electrodes and have demonstrated their ability to withstand oxidant environments while maintaining a good electrical conductivity and well electron-transfer-like surface, when a modified graphite felt is used as the positive electrode the energy density of the cell reaches values as high as  $13.84 \text{ W h kg}^{-1}$ , and the cell exhibits an excellent long-term performance with an energy density loss of only 6% after 3,000 galvanostatic charge–discharge cycles [272].

In cases where exist poor electric conductivity, GF are doping with ruthenium [273], nickel [274], and Mo [275]. Conductive graphite adhesive is typically used to support manganese oxides in an attempt to enhance conductivity [276]. For example, Mingping *et al.* used a polyaniline-MnO<sub>2</sub> graphite felt (PMGF) as a flexible electrode for supercapacitors, making a comparison with GF, highlighting that the specific capacitance of PMGF is much larger

than that GF [270]. PMGF is a promising electrode for supercapacitors because of GF facilitates the well dispersion of polyaniline-MnO<sub>2</sub> and for easy access of electrolytes.

The Ce<sup>3+</sup>/Ce<sup>4+</sup> redox couple in sulphuric acid has been employed to enhance the energy density of GF-based supercapacitors. This cell exhibits an excellent long-term performance with an energy density loss of only 6% after 3,000 galvanostatic charge–discharge cycles [272]. On the other hand, solid-state electrochemical capacitors, using GF, have given specific capacitances from 0.74 to 0.98 mF cm<sup>-2</sup>, together with a long cycle life and a short response time [277]. On the other hand, carbon fibre paper have recently been used as substrate material in supercapacitors containing several coatings such as nickel [278-280], nitrogen [281-284], silver [285] and manganese oxide [286-290], highlighting that these composites allow high energy densities (30-146 W h kg<sup>-1</sup>) and high retention capacities (85.2-98%) [282-284, 286, 290]. The use of GF with supported metallic catalyst as electrodes have demonstrated well performance in supercapacitors, maximizing significantly the capacitance of these devices. However, the application of GFs in supercapacitors remains limited.

## 5. Conclusions

Graphite felt provides a useful porous, 3-D electrode due its reasonable electrical conductivity, mechanical flexibility, compressibility and reasonable cost. The principal precursor used to manufacture the GF is PAN, which is a form of acrylic fibre and this type of precursor is stronger than other type of precursor-based carbon fibre. The carbonisation occurs at 700-1300 °C followed by graphitisation at 1600-1800 °C. GF shows a filamentary microscopic structure, with *ca.* 20 µm diameter fibres, exhibiting a high surface area, a

large porous volume and good conductivity being a useful electrode material. GF is chemically inert resistant to the most acids, alkalis and corrosive gases.

GF can be used in electrochemical reactor in flow-by, flow-through or flow-across configuration. The flow-through reactor provides a more compact volume, diminishing electrical resistance, pressure drops, and consequently low pump requirements. Meanwhile, the flow-by and flow-across configurations increase conversion efficiency but demands higher pump requirements than that obtained with the flow-through arrangement. The high conversion can be attributed to the increase of volumetric area exposed to the electrolyte.

The application of GF as anode or cathode in filter-press type cells permits considerable reduction in the stack size containing several cells, giving high power densities. This last is attributed to the extensive volumetric area, which allows high current intensities. It is worth mentioning that the high porosity of the GF leads to create turbulence, which enhances mass transport. This later helps to maintain acceptable potential and current distributions.

GF is used in several electrochemical applications, the principal use in electroanalysis is in sensors, identifying diverse metallic ions and chemical species owing to the high current intensity generated, which favors the limit of detection. GF is also doped with another materials or compounds obtaining composites that improve its performance for several applications such as electroanalysis, electrosynthesis, AOPS, RFBs, fuel cells, and super capacitors. Comparing the performance of GF versus other 3D carbon based materials, i.e. RVC, GF results better electrode material because of the extensive volumetric area, good conductivity and high porosity.

## 6. Recommendations for further research and development

1. The performance of well-defined GF materials should be quantified in terms of volumetric area, pressure drop and rate of mass transport of GF. Benchmarking against other carbon and metal electrodes is important.
2. There is a lack of clarity concerning changes in surface functionalisation due to pre-treatment of GF and there are opportunities for further Infra-red and Raman spectroscopy techniques, including *in-situ* studies.
3. The production of new structures of three-dimensional porous hybrid electrodes should be investigated. Such hybrid structures could combine the high surface area of inert graphite fibres and felts with mechanically rigid, supports such as carbon or metal meshes/foams.
4. In spite of GF electrode being well used in many electrochemical systems, some applications require further study, including energy conversion and storage, particularly in supercapacitors, where applications are scarce.
5. CFD simulations to characterise the reaction environment of GF electrodes are very limited and deserve increased attention; modelling and simulation are critical to optimise cell performance and develop novel electrochemical cells.

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**Table 1.** Volumetric surface area of some graphite felts.

| Company             | Type            | Volumetric surface area                |                             | Ref.  |
|---------------------|-----------------|----------------------------------------|-----------------------------|-------|
|                     |                 | Value / m <sup>2</sup> m <sup>-3</sup> | Method                      |       |
| SIGRI               | Sigratherm GFD5 | 24,000-60,000                          | Filamentary analogue-BET    | [291] |
| SIGRI               | Unknown         | 21,818                                 | Filamentary analogue        | [104] |
| Le Carbone-Lorraine | RVC 1000        | 31,000                                 | Filamentary analogue        | [88]  |
|                     | RVC 2000        | 150,000                                | Filamentary analogue        |       |
|                     | RVC 4000        | 23,500                                 | Filamentary analogue        |       |
|                     | CH (0.175)      | 11000                                  | Filamentary analogue        |       |
| Fibre Materials     | CH (0.25)       | 8,800                                  | Filamentary analogue        | [86]  |
| Le Carbone-Lorraine | RVC 4002        | 22100-22700                            | Ergun equation              | [11]  |
|                     |                 | 33,684                                 | Filamentary analogue        |       |
| Le Carbone-Lorraine | RVG 4000        | 18, 400                                | Kozeny-Carman/Kyan equation | [72]  |

**Table 2.**Chemical resistance of graphite [29].

| Chemical                                                                                                                                                           | Resistance to chemical attack |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| <i>Acids and Acid Solutions</i>                                                                                                                                    |                               |
| Mineral, non-oxidising (HCl, HCN, H <sub>3</sub> PO <sub>4</sub> , HF)                                                                                             | A                             |
| Boiling H <sub>2</sub> SO <sub>4</sub>                                                                                                                             | B                             |
| Mineral, oxidising (Br <sub>2</sub> , H <sub>2</sub> Cr <sub>2</sub> O <sub>7</sub> , HNO <sub>2</sub> , HClO <sub>4</sub> )                                       | B                             |
| Inorganic salts, acid-forming (Alum, BF <sub>3</sub> , CuCl <sub>2</sub> , NiCl <sub>2</sub> , sulfates)                                                           | A                             |
| Organic, strong (pH<3) (acetic, carbolic, formic, maleic, oxalic, phenol, picric, silicic)                                                                         | A                             |
| Organic, weak (pH 3 - 7) (abietic, benzoic, citric, cresol, lactic, palmitic)                                                                                      | A                             |
| Organic salts, acid-forming (allyl, amyl chlorides, ethyl chlorides)                                                                                               | A                             |
| <i>Alkalis and Alkaline Solutions</i>                                                                                                                              |                               |
| Mineral, non-oxidizing (hydroxides, NaOH, hydrazine, molten Mg, KOH)                                                                                               | A                             |
| Mineral, oxidizing (permanganate, perchlorate, perborates)                                                                                                         | B                             |
| inorganic salts, base forming (BaSH <sub>2</sub> , borax, phosphates, NH <sub>2</sub> O <sub>8</sub> , NaHSO <sub>2</sub> , NaPO <sub>3</sub> , Na <sub>2</sub> S) | A                             |
| Organic, strong (pH >11) (ethanol amines, pyridine)                                                                                                                | A                             |



| Chemical                                                                                                                                                                                                 | Resistance to chemical attack |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| Weak organic bases (pH 7-11) (aniline, soaps, urea)                                                                                                                                                      | A                             |
| <i>Gases</i>                                                                                                                                                                                             |                               |
| Acid (BF <sub>3</sub> , CO <sub>2</sub> , Cl <sub>2</sub> , HCl, HF, H <sub>2</sub> S)                                                                                                                   | A                             |
| Alkaline (wet NH <sub>3</sub> , steam to 300 °C)                                                                                                                                                         | A                             |
| Anhydrous (dew point below 0 °C) (NH <sub>3</sub> , CO <sub>2</sub> , ethane, F <sub>2</sub> , H <sub>2</sub> , HCl, HF, H <sub>2</sub> S, methane, O <sub>2</sub> to 150 °C, propane, SO <sub>2</sub> ) | B                             |
| Liquefied (air, F <sub>2</sub> , He, H <sub>2</sub> , methane, N <sub>2</sub> , O <sub>2</sub> )                                                                                                         | C                             |
| Oxidising (air above 250°C, F <sub>2</sub> , N <sub>2</sub> O <sub>4</sub> , O <sub>2</sub> above 150 °C, steam above 300 °C)                                                                            | C                             |
| Reducing (acetylene, ethane, methane)                                                                                                                                                                    | A                             |
| <i>Melts</i>                                                                                                                                                                                             |                               |
| Acid salts (AlCl <sub>3</sub> , H <sub>2</sub> BO <sub>3</sub> , FeCl <sub>3</sub> , PCL <sub>3</sub> , ZnCl <sub>2</sub> )                                                                              | B                             |
| Alkaline salts (Ba(OH) <sub>2</sub> , LiOH, KCN, soda ash)                                                                                                                                               | B                             |
| Metals (Al, Sb, Babbit metal, brass, Cu, Ga, Au, Mg, Hg, Ag, Sn, Zn)                                                                                                                                     | A                             |
| Neutral salts (KCl, Na <sub>2</sub> SO <sub>4</sub> )                                                                                                                                                    | A                             |
| Oxidising salts (sodium nitrate)                                                                                                                                                                         | B                             |

| Chemical                                                                                                                                                                                | Resistance to chemical attack |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| Salt solutions, neutral (baking soda, $\text{KCr}(\text{SO}_4)_2$ , $\text{CuSO}_4$ , $\text{Mg}(\text{SO}_4)_2$ , KCL, sea water, sewage, $\text{Na}_2\text{SO}_4$                     | A                             |
| <i>Solvents</i>                                                                                                                                                                         |                               |
| Aliphatic (butadiene, butane, butylene, cyclohexane, fuel oil, gasoline, lubric oil, propane, propylene)                                                                                | A                             |
| Aromatic (benzene, coal tar, creosote, cumene, naphtalene, petroleum, styrene, urethane)                                                                                                | A                             |
| Chlorinated, fluorinated ( $\text{CCl}_4$ , chlorobenzene, freons, chloroform, methyl chloride, vinyl chloride)                                                                         | A                             |
| Oxygenated, sulfide (acetaldehyde, acrolein, butyl acetate, butyl alcohol, $\text{CS}_2$ , rayon, ether, ethyl acetate, furfural, glycerin, methanol, ketones, sorbitol, vinyl acetate) | A                             |
| A = high, B = medium, C = low.                                                                                                                                                          |                               |

**Table 3.** Mass transport correlation for GF porous electrodes.

| Electroactive species    | Constants in the power law $k_m A_e = bu^c$ |       | Ref. |
|--------------------------|---------------------------------------------|-------|------|
|                          | b                                           | c     |      |
| $\text{Fe(CN)}_6^{3-}$   | 0.019                                       | 0.352 | [86] |
| $\text{Fe(CN)}_6^{3-}$   | 0.0012                                      | 0.61  | [87] |
| $\text{Hg}^{+2}$         | 3.7                                         | 0.42  | [88] |
| $\text{Fe(CN)}_6^{3-}$   | 6.0                                         | 0.36  | [88] |
| $\text{Cu}^{2+}$         | 3.38                                        | 0.62  | [5]  |
| $\text{Na}_2\text{SO}_4$ | 0.0012                                      | 1.56  | [76] |
| $\text{NaClO}_4$         | 0.0028                                      | 1.47  | [76] |

**Table 4.** Examples of recent sensor studies using GF electrodes.

| Species                       | Concentration /<br>mol dm <sup>-3</sup> | Flow geometry | Technique | Ref.  |
|-------------------------------|-----------------------------------------|---------------|-----------|-------|
| Ag <sup>+</sup>               | 2.5×10 <sup>-8</sup>                    | Quiescent     | ASV       | [122] |
| Zn                            | 5×10 <sup>-7</sup>                      | Flow cell     | SV        | [127] |
| Pb <sup>+2</sup>              | 1×10 <sup>-9</sup>                      | Flow Through  | LSSV      | [128] |
| Pb <sup>+2</sup>              | 2.5×10 <sup>-8</sup>                    | Flow Through  | LSSV      | [129] |
| Ag <sup>+</sup>               | 1-3×10 <sup>-8</sup>                    | Quiescent     | DPASV     | [126] |
| Hydrazine                     | 4.9×10 <sup>-7</sup>                    | Stirring      | AD        | [130] |
| Nitrite                       | 1.3×10 <sup>-8</sup>                    | Quiescent     | CV        | [131] |
| H <sub>2</sub> O <sub>2</sub> | 1.2×10 <sup>-9</sup>                    | Quiescent     | AD        | [132] |

Anodic stripping voltammetry (ASV), swept voltammetry (SV), linear swept stripping voltammetry (LSSV), differential pulse anodic stripping voltammetry (DPASV), anodic dissolution (AD), and cyclic voltammetry (CV)

**Table 5.** GF as an electrode material in RFBs.

| Electrode material                                                            | Manufacturer            | Thickness /mm | Electrode polarity | RFB system          | Ref.  |
|-------------------------------------------------------------------------------|-------------------------|---------------|--------------------|---------------------|-------|
| GF GFA-type                                                                   | SGL Group, Germany      | 8             | +ve                | Zinc-cerium         | [292] |
| GF                                                                            | Le Carbone, France      | *             | +ve and -ve        | All-vanadium        | [292] |
| PAN-based graphite felt                                                       | Xin Xing Carbon, China  | 5             | +ve                | Bromine-polysulfide | [55]  |
| Cobalt coated PAN-based graphite felt                                         | Xin Xing Carbon, China  | 5             | -ve                | Bromine-polysulfide | [55]  |
| Graphite felt bonded electrode assembly with non-conducting plastic substrate | FMI Graphite, USA       | *             | +ve and -ve        | All-vanadium        | [293] |
| Graphite felt                                                                 | COS1011, CeTech, Taiwan | 6             | +ve and -ve        | All-vanadium        | [225] |
| Graphite felt – GFA series                                                    | SGL company             | 1.5           | +ve and -ve        | All-vanadium        | [244] |

\* Not given.

**Table 6.** Electrode type, configuration, electrode size, inoculation source, reactor configuration maximum current density of GF in MFC.

| Electrode                    | Configuration               | Electrode size                          | Inoculation source                                                | Reactor configuration                                            | Maximum volumetric power | Ref.  |
|------------------------------|-----------------------------|-----------------------------------------|-------------------------------------------------------------------|------------------------------------------------------------------|--------------------------|-------|
| GF - Anode                   | Packed                      | Anode chamber:<br>156 cm <sup>3</sup>   | Bacteria from an active MFC                                       | Two chamber MFC<br>catholyte: K <sub>3</sub> Fe(CN) <sub>6</sub> | 386 W m <sup>-3</sup>    | [294] |
| GF - Cathode                 | Continuous MFC              | 8 cm × 4 cm × 0.4 cm                    | Sediment sample, and culture<br>obtained from MFC on acetate      | Four cell connected<br>into one block                            | 37 W m <sup>-3</sup>     | [295] |
| GF - Cathode                 | Flow-through<br>mini-stacks | 20 × 10 <sup>-3</sup> m <sup>2</sup>    | R. ferrireducens strain T118 <sup>T</sup>                         | Two chambered glass<br>vessel                                    | 0.2275 W m <sup>-3</sup> | [266] |
| GF - Anode                   | Cassette electrodes<br>MFC  | 3 cm × 20 cm × 0.15<br>cm               | Synthetic wastewater containing<br>stach, petone and fish extract | 12 chamber CE-MFC                                                | 129 W m <sup>-3</sup>    | [296] |
| GF                           | Flow-through                | 1.2 cm <sup>3</sup>                     | Strain of Shewanella oneidensis<br>DSP10                          | Two chamber<br>mini-MFC                                          | 500 W m <sup>-3</sup>    | [297] |
| GF                           | Plane                       | 7 cm <sup>2</sup>                       | Anaerobic sludge                                                  | Dual chamber                                                     | 2.6 W m <sup>-3</sup>    | [269] |
| GF                           | Plane                       | Cathode chamber:<br>110 cm <sup>3</sup> | Aerobic/anaerobic sludge                                          | Dual chamber<br>Flat plate MFC                                   | 14.1 W m <sup>-3</sup>   | [298] |
| Manganese<br>oxide coated-GF | Tubular                     | Cathode chamber:<br>40 cm <sup>3</sup>  | Mixture of sediment, aerobic<br>and anaerobic sludge              | Cylindrical dual<br>chamber with an<br>internal anode chamber    | 83 W m <sup>-3</sup>     | [299] |

**Figure captions**

- Figure 1      Examples of porous, 3-D electrodes in carbon and metals.  
a) graphite felt, b) granular C particles, c) woven C cloth and d) glassy C foam (100 ppi RVC); e) Ti felt, f) Ti micromesh, g) Ti expanded mesh and h) 3-D laser printed stainless steel filamentous mesh.
- Figure 2      Modes of operation of GF with respect to current and electrolyte flow.  
a) flow-by. b) flow-through and c) flow-across.
- Figure 3.      Timeline of applications using graphite felt as an electrode.
- Figure 4.      The relationships among structure, properties and applications of GF electrodes.
- Figure 5.      The diversity of fibre and felt structures of graphite felt (GF) and its common polyacrylonitrile (PAN) or Lycra (polyester/polyurethane) precursor.
- Figure 6.      SEM micrographs of the GF taken at various magnifications for two different sheet orientations. a)  $\times 20$ , b)  $\times 100$ , c)  $\times 500$ , d)  $\times 5000$  (cross-section) e)  $\times 20$ , f)  $\times 50$  [11].

- Figure 7. Schematic representation of the main types of precursor used during the manufacture of graphite felt.
- Figure 8. Electrochemical cells: (a) GF electrode in the flow-through configuration for silver removal [71]. (b) GF electrode in the flow-by configuration used for the electrochemical reduction of Cr (VI) (**A**: anodic compartment and **C**: cathodic compartment) [72]. (c) GF electrode in the flow-through configuration used for metal ion removal from wastewater [75].
- Figure 9. (a) Influence of electrolyte flow rate on the RTD curve, and (b) dependence of axial dispersion coefficient with flow rate, in the GF composite [81].
- Figure 10. (a) Pressure drop over the CNF composite for a water-ethanol mixture: (●) water, (○) 10% ethanol, (▼) 20% ethanol, (Δ) 30% ethanol, (■) 40% ethanol, (□) 50% ethanol, (◆) 60% ethanol, (◇) 70% ethanol, (▲) 80% ethanol, (▽) 95% ethanol. (b) Changes in pressure drop and permeability coefficient with ethanol concentration at a mean linear flow velocity of  $0.01 \text{ m s}^{-1}$  [80]. (c) Mass transport coefficients and (d) experimental potential distribution as a function of mean linear flow velocity in a flow-through reactor, indicating the potential region where hydrogen evolution occurs [5].
- Figure 11. Normalized concentration profiles on a 5 mm long porous electrode. After Doherty *et al.* [108].



Figure 12. Local distributions of (a) concentration, (b) overpotential, and (c) current density in the negative half-cell corresponding to averaged electrolyte flow velocities of  $0.264 \text{ mm s}^{-1}$  with an inlet SOC of 50% during galvanostatic discharge at  $400 \text{ A m}^{-2}$ . After Qiu *et al.* [109].

Figure 13. Comparison between graphite felt (GF) and reticulated vitreous carbon (RVC) cathodes used in the flow-by cell for mediated electrosynthesis of  $\text{H}_2\text{O}_2$  at pH 3. Organic phase: TBP –  $0.2 \text{ mol dm}^{-3}$  EtAQ +  $0.1 \text{ M}$  TBAP, aqueous phase:  $1 \text{ mol dm}^{-3}$   $\text{Na}_2\text{SO}_4$  at pH 3 +  $1 \times 10^{-3} \text{ mol dm}^{-3}$  A336 per emulsion. Phase volume ratio:  $2.1/1_{\text{aq/org}}$ . Emulsion mass flux:  $4.25 \text{ kg m}^{-2} \text{ s}^{-1}$ ;  $\text{O}_2$  mass flux  $0.026 \text{ kg m}^{-2} \text{ s}^{-1}$  (at STP),  $0.1 \text{ MP}$  and  $308 \text{ K}$ . Legend: (a)  $\text{H}_2\text{O}_2$  concentration per pass, (b) current efficiency for  $\text{H}_2\text{O}_2$ . Electrode material: (■) graphite felt, (●) 39 ppi reticulated vitreous carbon, (▲) 12 ppi RVC. After Gyenge and Oloman [138].

Figure 14 Electrodeposition of tin-copper alloys a) on a porous, 100 ppi reticulated vitreous carbon support having 97% volumetric porosity; b) cross-sectional view of the RVC; (c) GFA5 (Sigrafelt) carbon felt (volumetric porosity of 68%); (d) a higher magnification view of deposits from b), showing single strands of the felt having a typical diameter of  $5\text{-}10 \mu\text{m}$ .

Figure 15. The effect of current density on the kinetics of degradation of SMT using the GF/GF electrode couple: for all (a) electro-Fenton, (b) anodic oxidation  $V_s =$

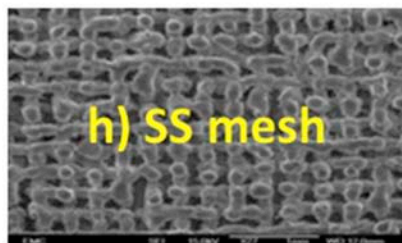
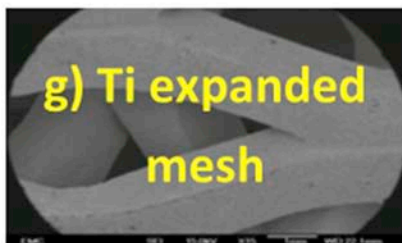
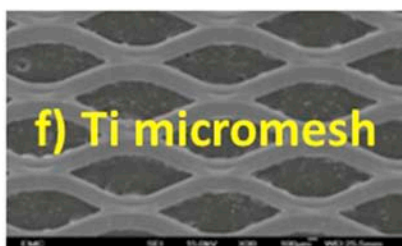
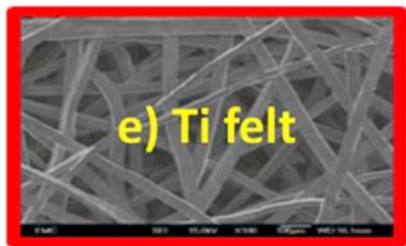
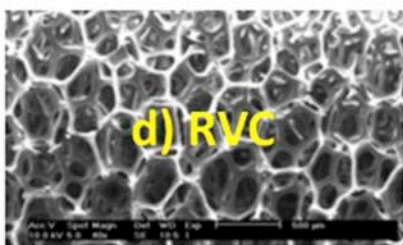
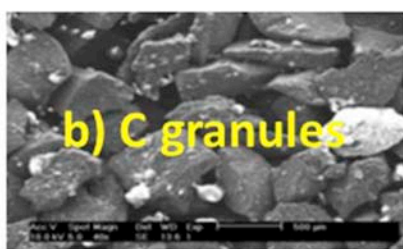
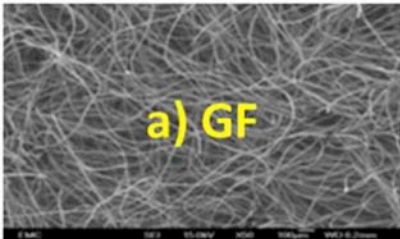
300 cm<sup>3</sup>, pH 3, [Na<sub>2</sub>SO<sub>4</sub>] = 50 mmol dm<sup>-3</sup>, [Fe<sup>2+</sup>] = 0.2 × 10<sup>-3</sup> mol dm<sup>-3</sup> (for electro-Fenton only) [166].

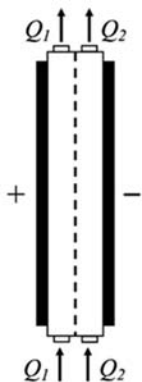
Figure 16. Copper ion removal from synthetic drainage water, influence of the solution flow rate and current density for a graphite felt cathode. Open symbols: electrolyte evacuated and kept under nitrogen atmosphere during electrolysis; full symbols: electrolysis in an open atmosphere [173].

Figure 17. Cycling tests of VRFBs using electrolyte with and without 0.005 mol dm<sup>-3</sup> Cu<sup>2+</sup> ions. (a) EE and (b) discharge capacity at a current density of 200 mA cm<sup>-2</sup> on the GF electrode [244].

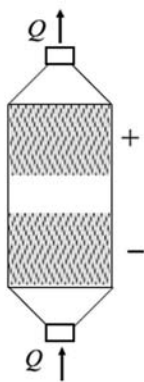
Figure 18. Comparison of the maximum power densities in two similar applications of GF in H<sub>2</sub>/Br<sub>2</sub> fuel cell using GF modified by acidic oxidation and thermal oxidation [251].

Figure 19. Power density and cell polarisation curves for an MFC equipped with graphite felt (GF), carbon paper (CP) or stainless steel mesh (SSM) as a biocathode [269].

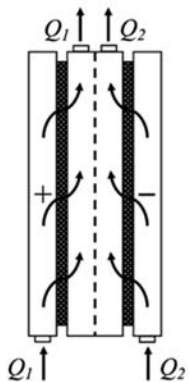




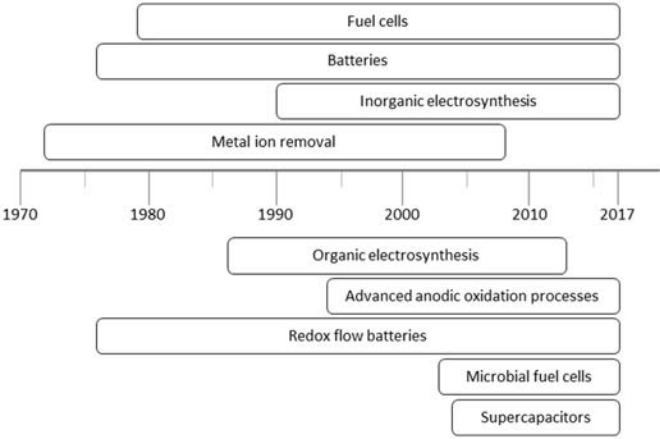
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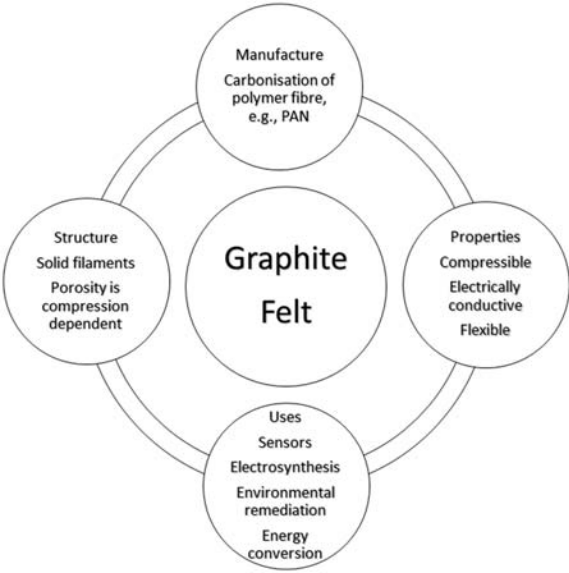


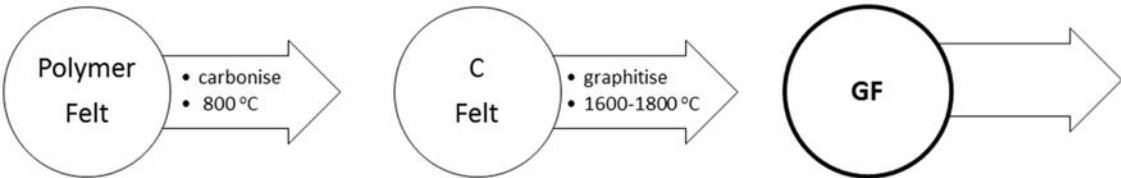
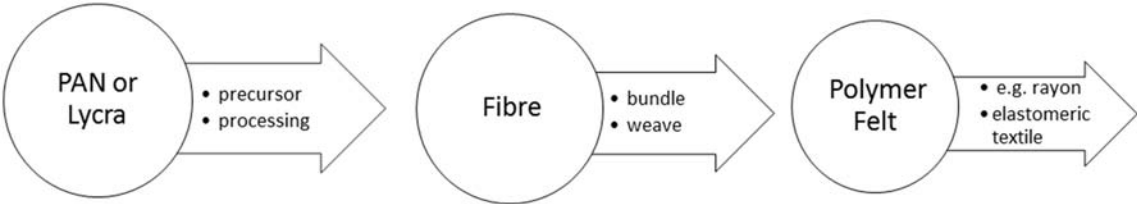
b) Flow-through

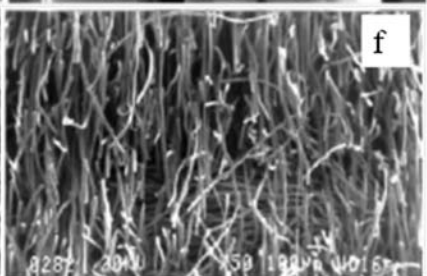
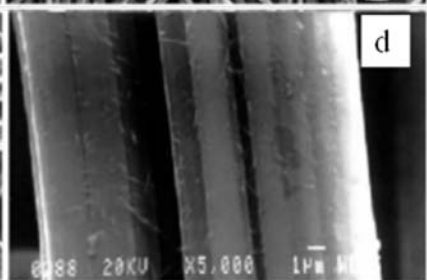
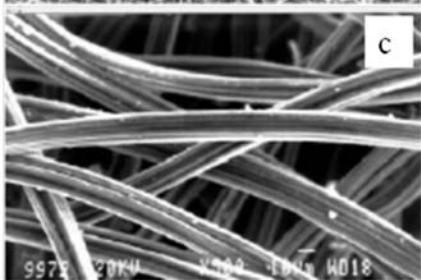
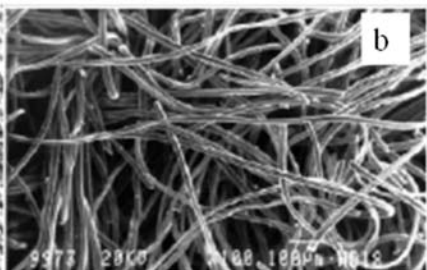
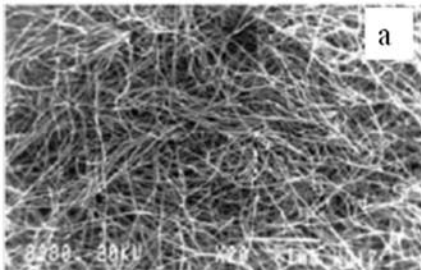


c) Flow-across

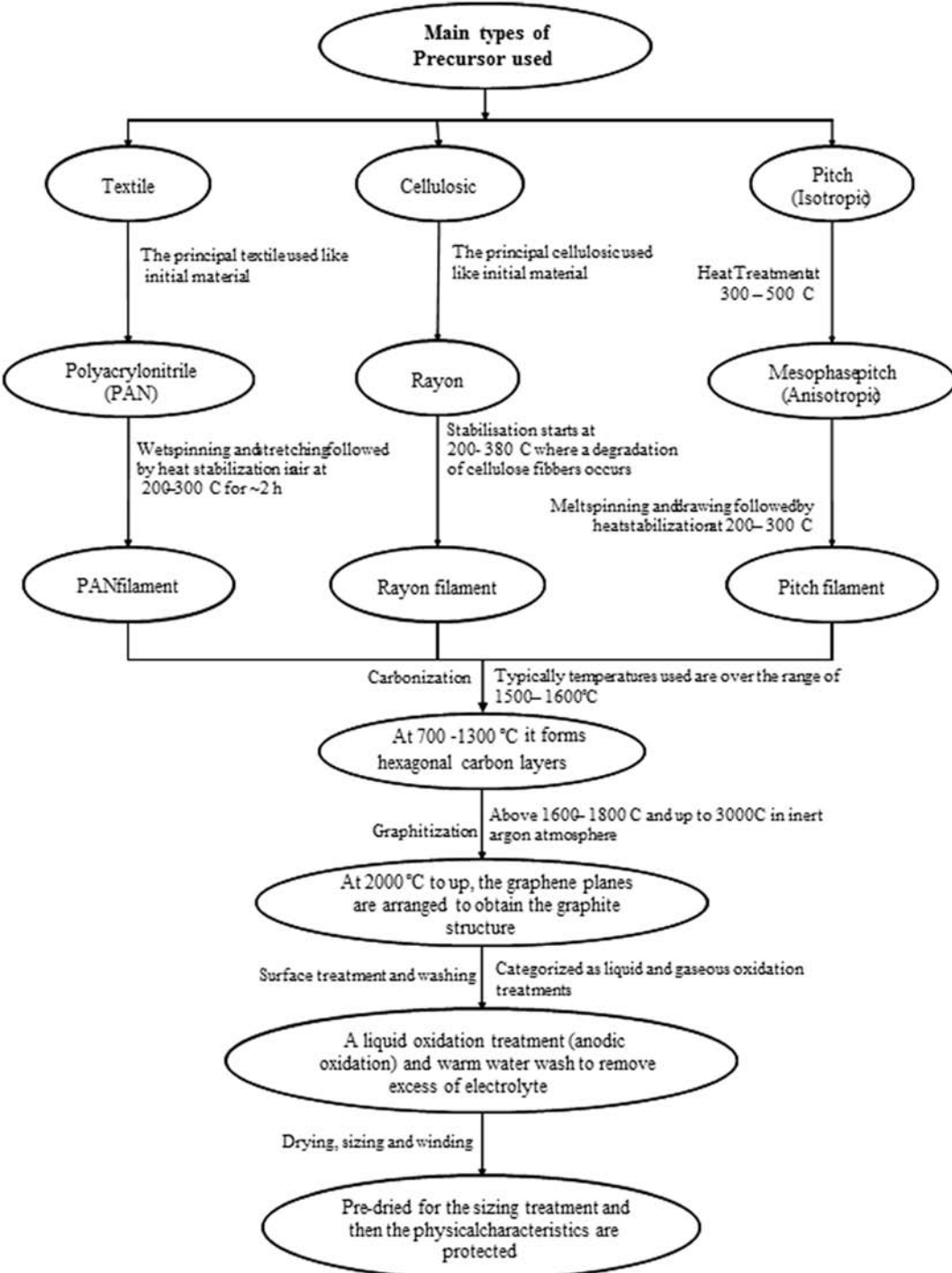




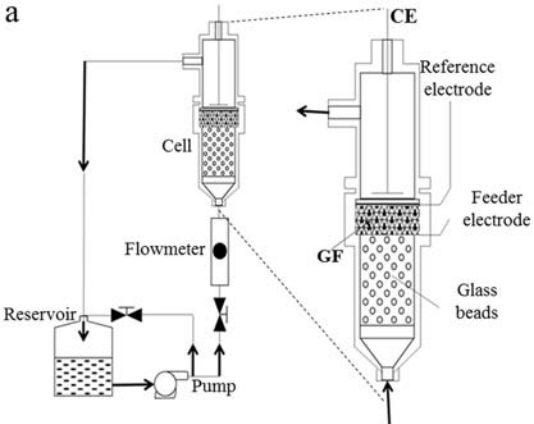




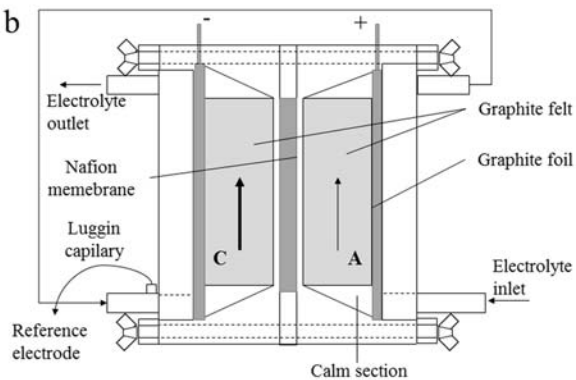




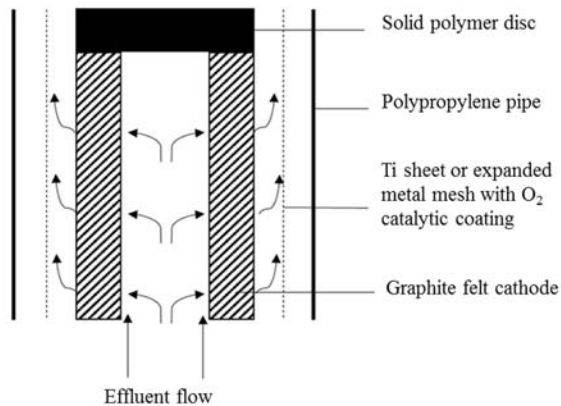
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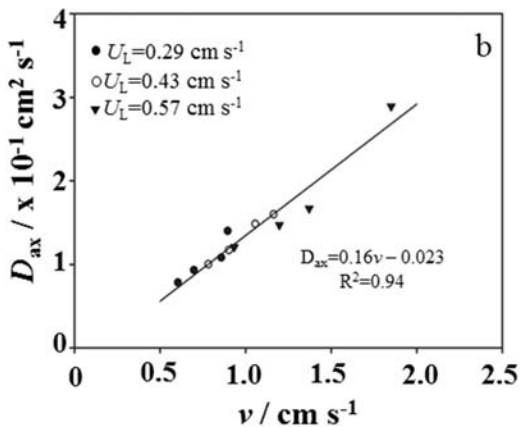
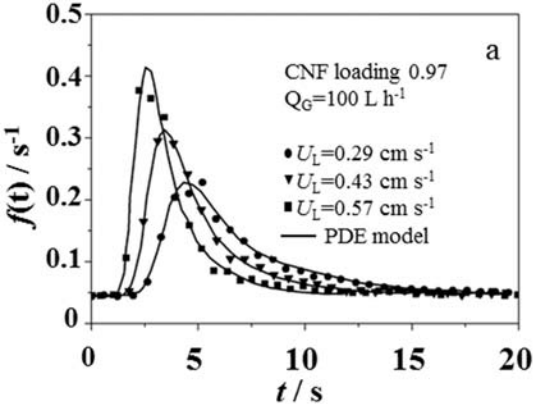


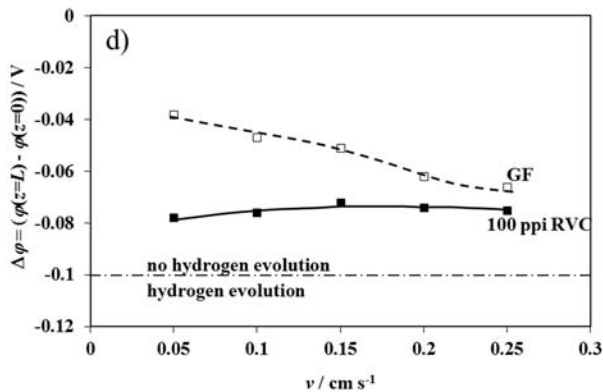
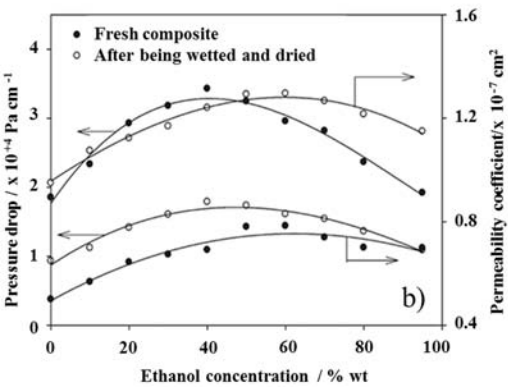
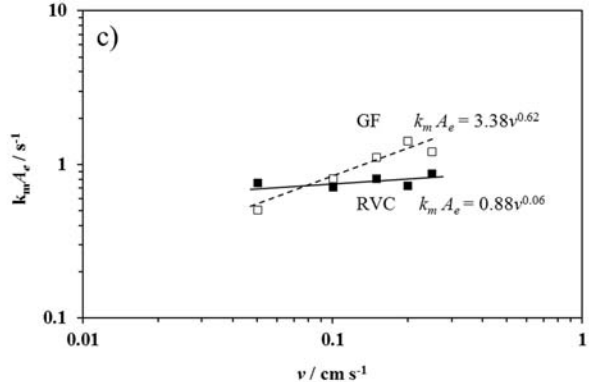
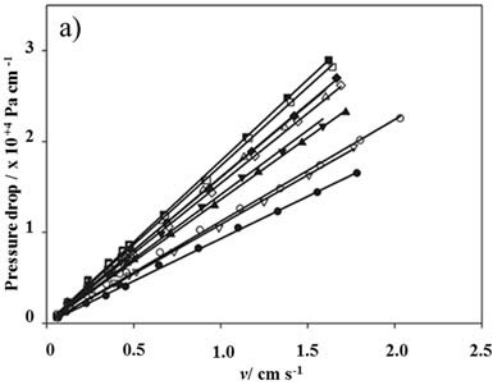
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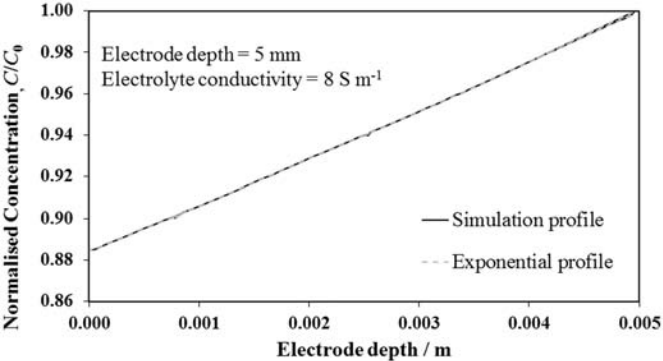


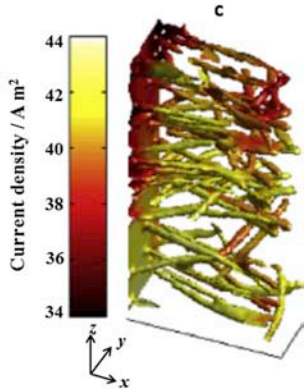
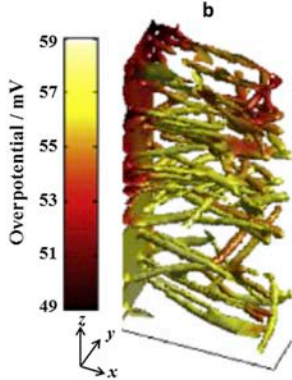
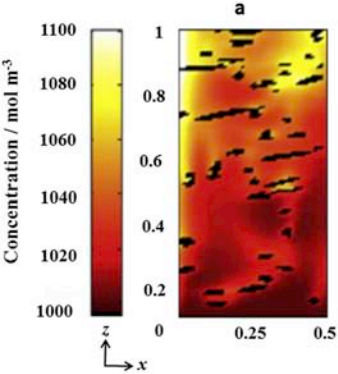
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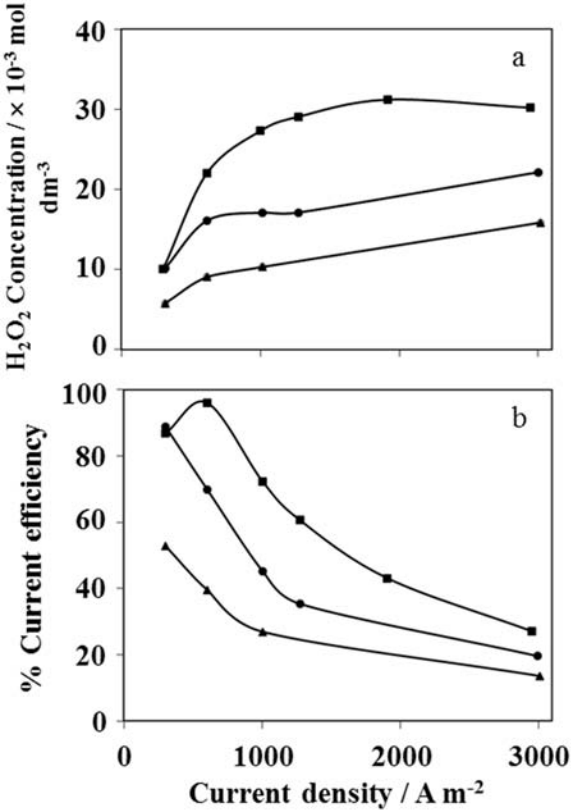


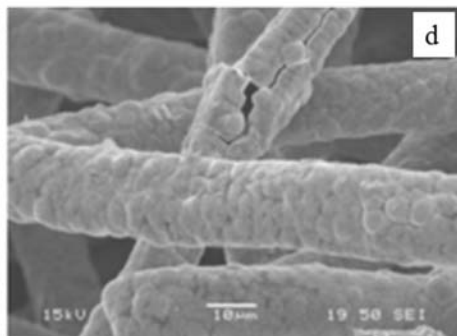
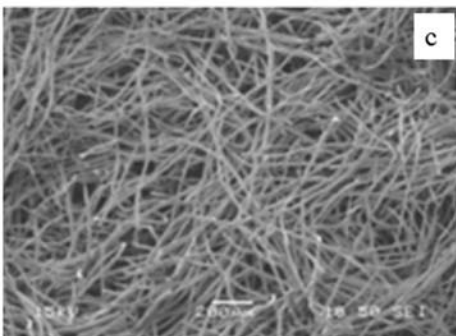
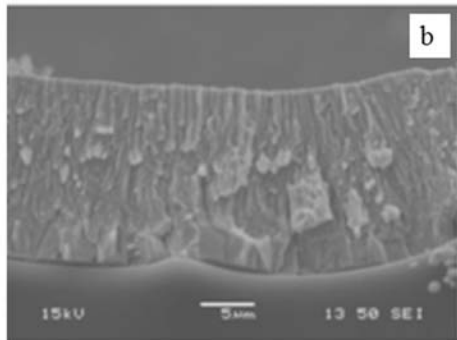
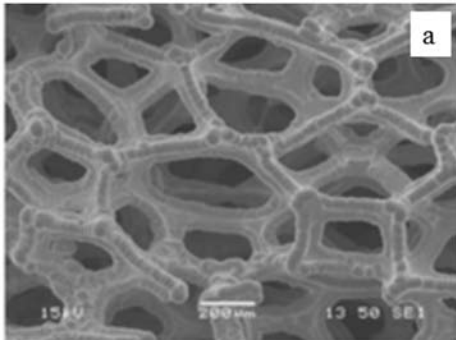




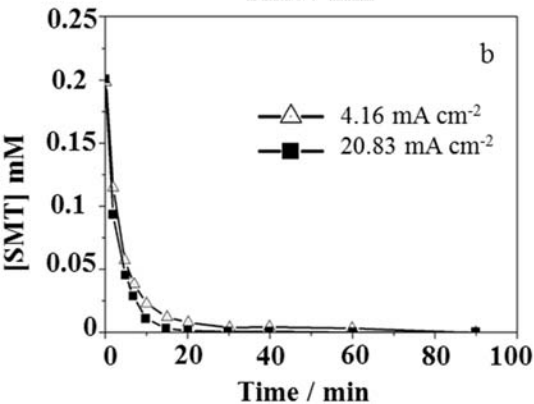
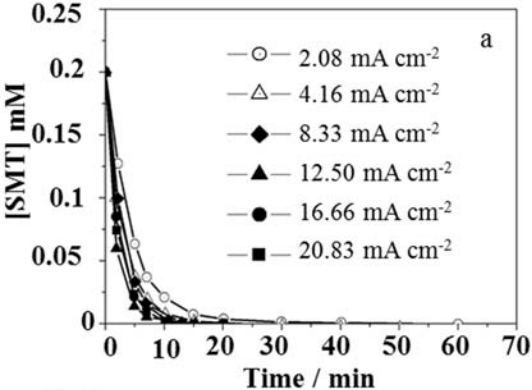


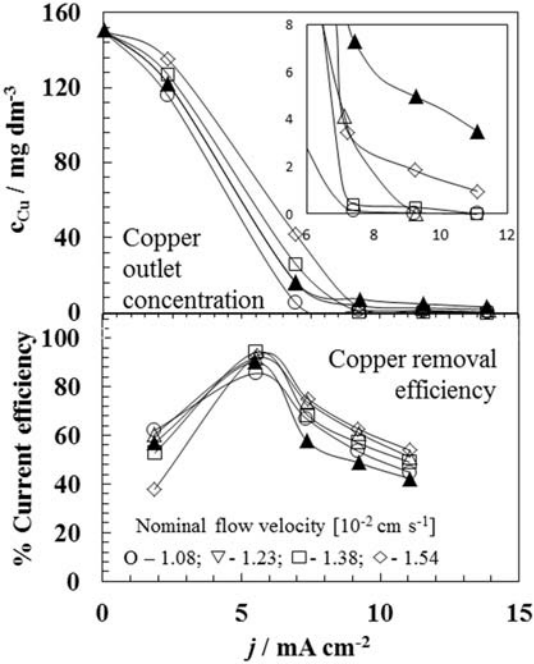


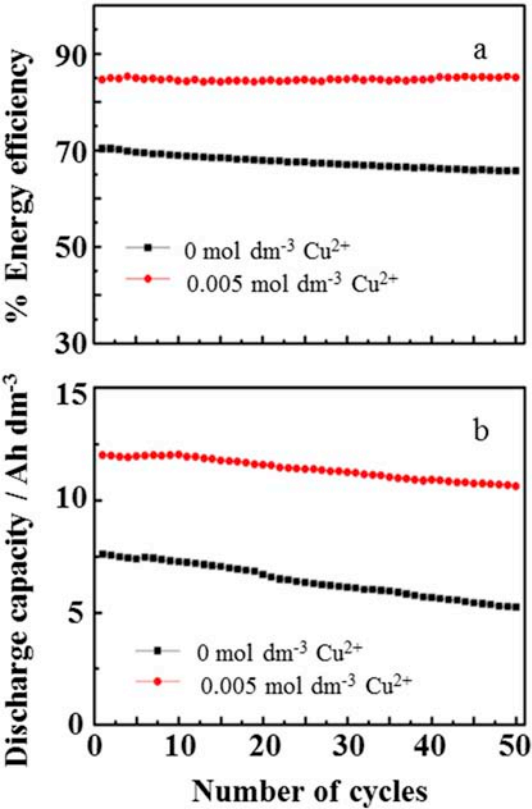


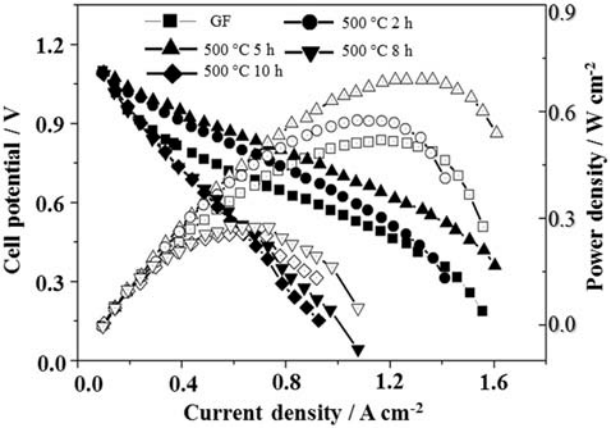


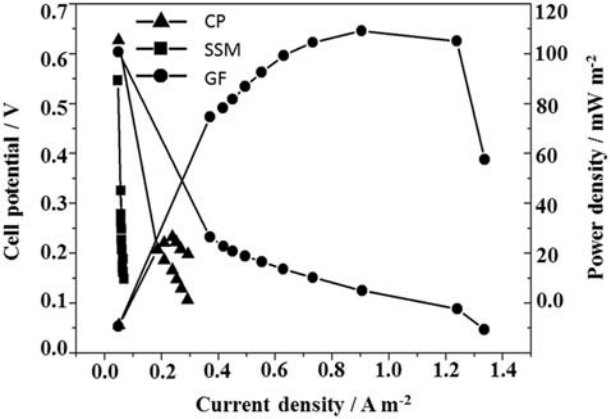












**Highlights**

- Chemical, physical and electrical characteristics of GF enable diverse uses
- GF shows reasonably high overpotentials for oxidation and reduction of water
- The texture of GF felt can act as a turbulent promoter to enhance mass transport
- A high space time yield per pass in electrochemical reactors can be achieved
- Applications span energy, electrosynthesis, electrodeposition and water purification